

Significance

Clinical decisions often require collaboration among specialists who must rapidly integrate diverse expertise and current evidence. Existing AI systems rarely replicate this workflow or provide transparent sourcing. MedAgents-X bridges this gap by coordinating small, role-diverse teams of language model “experts” under a moderator who directs discussion and evidence retrieval. This framework unites adaptive team assembly, structured facilitation, and hybrid retrieval across medical literature and web sources to produce consensus recommendations with verifiable citations. In blinded clinician evaluations, MedAgents-X generated more appropriate and actionable answers with improved citation fidelity. These findings suggest a viable path toward AI systems that support clinical reasoning in a transparent, accountable, and defensible manner.

MedAgents-X: A Multi-Agent Framework for Evidence-Based Medical Reasoning

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Abstract

Clinical decision-making depends on synthesizing evidence and coordinating input from multiple specialists—capabilities that most current AI systems lack, particularly with verifiable sourcing. MedAgents-X addresses this gap through moderated, evidence-grounded collaboration among role-specialized language model “experts.” A triage process assembles small, diverse teams; each member retrieves relevant literature and web-based evidence; and a moderator guides focused, multi-round discussion to reach a cited consensus. Evaluated on nine challenging medical question-answering benchmarks deliberately enriched for difficult, unsolved cases, MedAgents-X achieves substantial gains over strong single-agent and retrieval-only baselines (e.g., +26.3 percentage points in mean accuracy with o3-mini) while remaining computationally efficient. Ablation studies identify a practical operating point—three experts, two discussion rounds, and chaired facilitation—that mirrors effective clinical consultation patterns. Blinded evaluations by practicing clinicians show higher clinical appropriateness, greater “would-act” agreement, and improved citation fidelity. Together, these results demonstrate that orchestrated, evidence-governed multi-agent collaboration enhances both the reliability and transparency of AI-assisted medical reasoning, offering a feasible blueprint for accountable clinical decision support.

Keywords: medical artificial intelligence | multi-agent systems | clinical decision support | evidence-based medicine | large language models | retrieval-augmented generation | collaborative AI | healthcare informatics

Introduction

Clinical decision-making requires accurate reasoning, timely access to current evidence, and coordinated input from multiple specialties. Medical errors affect one in ten patients globally; in the United States alone, diagnostic errors contribute to tens of thousands of deaths annually [1]. Contributing factors include cognitive biases, fragmented evidence synthesis, and limited access to the right expertise at the right time. Clinicians mitigate these risks by integrating literature with domain knowledge and, for difficult cases, convening multidisciplinary consultations and tumor boards.

Conventional clinical decision support systems (CDSS) [2, 3] rarely match this standard: they may retrieve information or reason well, but seldom deliver both with the depth and traceability required for high-stakes decisions. Recent large language models (LLMs) and agent systems show strong performance on medical tasks [4–12], yet hallucination of clinical facts remains a critical barrier [13, 14]. Retrieval-augmented generation (RAG) grounds responses in sources [15, 16], advancing beyond keyword lookups to corpus- and graph-aware retrieval [17–19]. In parallel, multi-agent frameworks such as MAC [20], MedAgents [21], and MDAgents [22] emulate collaborative practice and can outperform monolithic models. A clinically salient question remains: can collaborative, retrieval-augmented systems produce verifiable, evidence-grounded recommendations suitable for decision support?

We present MedAgents-X, a framework that operationalizes evidence-grounded, multi-agent deliberation for medical reasoning. MedAgents-X (i) assembles a small panel of role-specialized experts via adaptive triage, (ii) equips each expert with medical evidence retrieval over $\sim 2.45\text{M}$ curated segments plus web search, and (iii) orchestrates a focused, moderator-guided discussion to synthesize a consensus answer with citations. Three design choices anchor the approach: a hierarchical search stack for coverage and recency, adaptive triage that matches specialties and discussion depth to case difficulty, and discussion modes that trade off thoroughness and efficiency. The objective is an AI system that integrates collaborative reasoning with rigorous retrieval and provides verifiable evidence and transparent deliberation.

Unlike prior frameworks such as *MedAgents* [21] and *MDAgents* [22], which demonstrated the feasibility of agent collaboration in zero-shot medical reasoning, MedAgents-X redefines this paradigm around verifiable, evidence-grounded deliberation. MedAgents-X shifts from outcome-focused coordination to process-level governance of evidence and interaction, marking a transition from collaboration to accountable deliberation. Rather than coordinating independent agents through heuristic voting, MedAgents-X introduces a moderator–clinician–triage architecture that operationalizes core elements of real-world consultation: adaptive team assembly, structured facilitation, and explicit evidence arbitration. This reframing

treats facilitation not merely as orchestration, but as a form of epistemic regulation—governing how agents generate, contest, and reconcile evidence during collective reasoning.

This work makes three contributions. First, MedAgents-X is the first multi-agent RAG system designed specifically for medical reasoning, integrating adaptive triage, role-specialized deliberation, and explicit evidence governance with auditable traces. The system achieves state-of-the-art accuracy while maintaining clinical efficiency. Second, systematic ablations yield workflow insights that mirror multidisciplinary consultations—small, role-diverse teams with chaired facilitation, two focused rounds, and explicit evidence checks—offering concrete guidance for multi-agent system design in medicine. Third, blinded human evaluation demonstrates higher clinical appropriateness and *would-act* rates, alongside improved citation fidelity, providing independent evidence that retrieval quality and orchestration together strengthen evidentiary defensibility.

We evaluate MedAgents-X on nine medical QA tasks [23–30], focusing on a curated hard subset where many questions remain unsolved by state-of-the-art systems [31]. MedAgents-X achieves state-of-the-art mean accuracy ($\sim 32\%$) and consistently outperforms strong single-agent, retrieval-only, and multi-agent baselines. Systematic ablations show that retrieval, orchestration, multi-agent collaboration, iterative refinement, and role specialization each contribute complementary gains. Finally, MedAgents-X produces transparent, citation-grounded rationales and auditable traces to support accountability and clinical usability.

Results

Overview of MedAgents-X

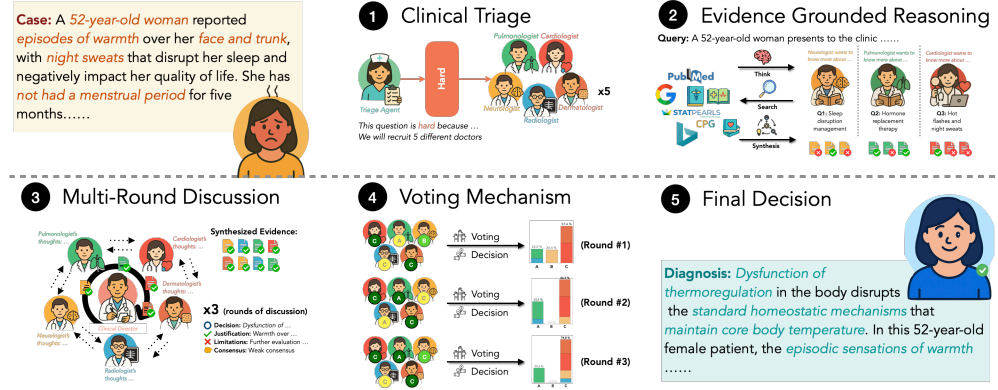


Fig. 1 The MedAgents-X framework. MedAgents-X operates through five sequential phases: clinical triage to assess question difficulty, evidence-grounded reasoning by specialized clinicians, multi-round discussion to resolve conflicts, vote-based aggregation of clinician opinions, and final decision with supporting citations.

We developed MedAgents-X, an evidence-grounded, multi-agent clinical reasoning framework that brings together specialized medical experts to answer complex healthcare questions. The system draws from 49 different medical specialties and uses a carefully curated library of approximately 2.45 million medical sources that can be cited and referenced. When needed, it can also search trusted medical websites for the most recent information. This is a large-scale system in breadth of expertise, transparency of sourcing, and rigor of evaluation across nine challenging medical QA tasks. Three core design principles make the system work effectively: carefully curated questions and evidence, smart collaboration between agents, and an evidence-first retrieval-to-decision workflow (Fig. 1).

We assemble a hard evaluation pool by focusing on medical questions that even strong AI systems struggle to answer. Questions are balanced across different medical topics and formats, checked for potential data contamination, and reviewed by practicing clinicians to ensure they require multi-step reasoning and careful discrimination between options. For our evidence base, we constructed a high-precision medical library from clinical guidelines, point-of-care reviews, textbooks, and other authorized sources. Duplicates are removed, different versions of documents are tracked, and medical terminology is standardized throughout. Sensitive content is filtered out, and every document can be cited appropriately. To stay current with medical advances, clinicians can search the web during consultation to find recent guidance, such as new clinical trials or updated society recommendations. This new information only influences decisions when explicitly cited in the final answer.

The process begins with a quick assessment of the question’s difficulty, which helps assemble a small, diverse panel of clinician agents whose specialized knowledge guides the entire consultation. Each clinician identifies what information is still needed, clarifies medical terminology and abbreviations, and searches our curated library for supporting evidence. The system cross-checks findings and removes redundant or overly general information. The clinicians then draft their reasoning with direct links to specific citations. A brief moderated discussion follows, where the system identifies areas of agreement, conflict, and information gaps. This focuses the next round of evidence-gathering on information that could actually change the final decision. The consultation typically reaches a conclusion within one or two focused rounds, producing a single answer with complete references and a concise audit trail that clinicians can review.

MedAgents-X establishes a new state-of-the-art in nine medical reasoning benchmarks

Many clinically important diagnostic decisions require complex reasoning that current AI systems struggle to support effectively. Tasks such as interpreting conflicting diagnostic evidence typically require expert clinicians to synthesize information from multiple sources, weigh competing hypotheses, and reach defensible conclusions. This process is both cognitively demanding and difficult for single AI agents to replicate reliably. No current foundation model can consistently match the

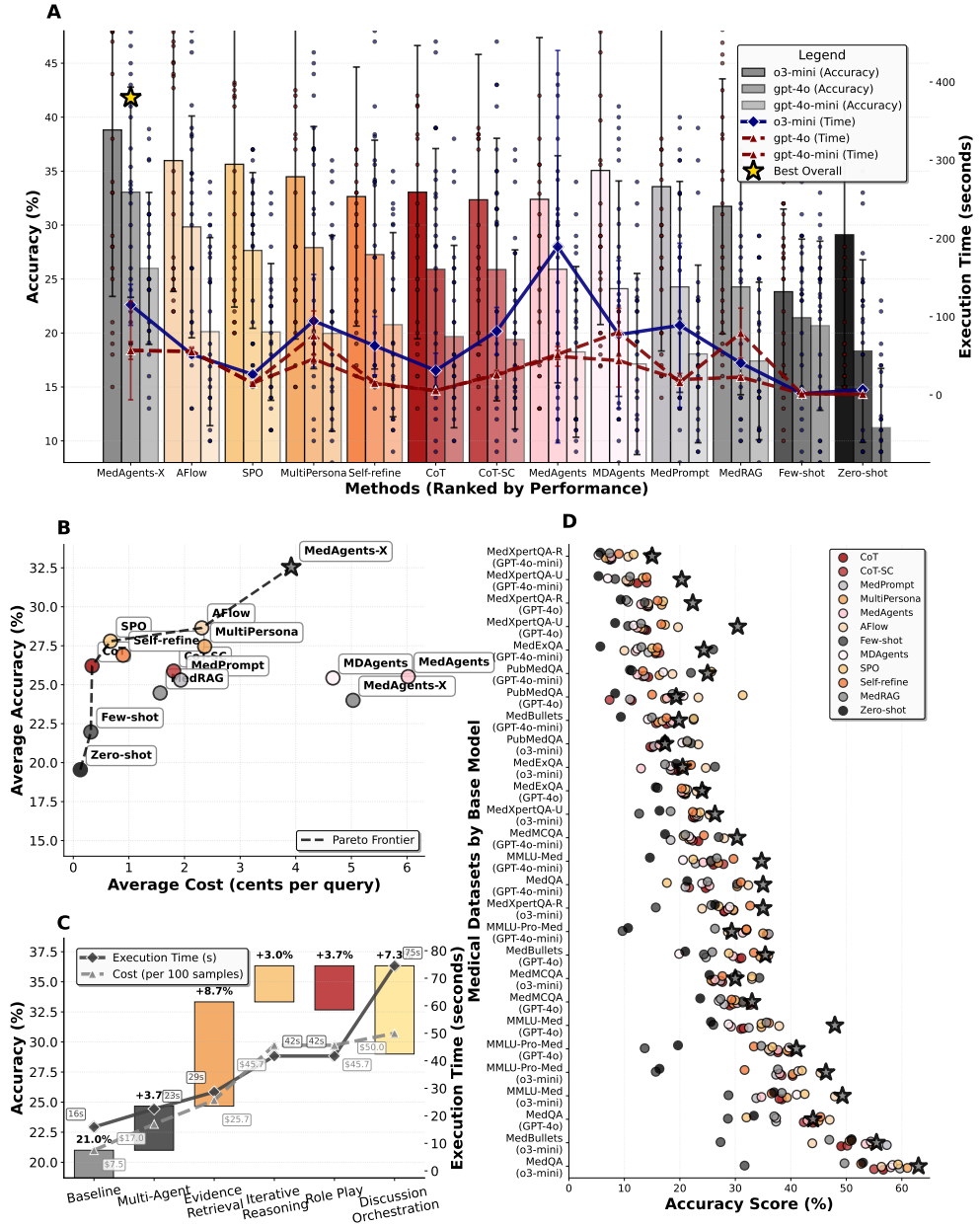


Fig. 2 Overall performance and efficiency of MedAgents-X. **(a)** With o3-mini, MedAgents-X attains 32.4% average accuracy and outperforms twelve strong baselines (Table A1). Bars show accuracy, and the line shows execution time. **(b)** Quality-resource frontier. MedAgents-X lies on the Pareto front with a cost of \$0.004 per query. **(c)** Efficiency decomposition showing improvements from multi-agent routing and retrieval over single-agent baselines. **(d)** Improvements occur across clinical, research, and reasoning tasks, and collaborative evidence-based deliberation transfers across different epistemic settings.

collaborative reasoning that emerges when multiple specialists deliberate over challenging cases. Moreover, deploying individual AI assistants for each diagnostic task would lack the cross-specialty perspectives and evidence synthesis that characterize high-quality clinical decision-making. We evaluated MedAgents-X across nine challenging medical reasoning benchmarks, comparing against twelve state-of-the-art baselines to assess whether orchestrated multi-agent collaboration can match the performance of expert clinical teams (Fig. 2).

MedAgents-X establishes a new state-of-the-art across all benchmarks, achieving 32.4% mean accuracy with o3-mini, a 26.3-point absolute gain over zero-shot prompting and surpassing the next-best method by over five percentage points (Fig. 2a). The consultation paradigm of triage, specialized expertise, and evidence-grounded deliberation addresses fundamental limitations that simple prompting cannot overcome. These advantages persist across model families: even with smaller models like gpt-4o-mini, MedAgents-X outperforms stronger baselines running on larger models, demonstrating that architectural innovation complements rather than merely exploits model capacity. In clinical settings, diagnostic support systems face constraints like limited specialist time, evidence retrieval, and deliberation overhead. We treat cost as a resource utilization proxy, not just runtime. MedAgents-X achieves a good quality–resource balance at \$0.004 per query, remaining Pareto-efficient (Fig. 2b). No other method reaches similar quality without higher deliberation costs. This efficiency comes from adaptive triage, which adjusts consultation depth to case difficulty, mimicking how teams intensify discussion when uncertainty is high.

To understand what allows MedAgents-X to make better decisions, we systematically removed each stage of the consultation process. Retrieval improved results most (+8.7%) by grounding decisions in verifiable evidence; orchestration (+7.3%) made sure disagreements were resolved through targeted follow-up; multi-agent collaboration (+3.7%) revealed competing hypotheses; iterative refinement (+3.0%) allowed belief updates; and role specialization (+1.7%) focused search on relevant clinical distinctions (Fig. 2c). These gains, roughly additive, show each part addresses different reasoning failures, not redundant improvements.

We also show that these improvements are not limited to a single problem family (Fig. 2d). Tasks grounded in real-time diagnostic reasoning (MedQA, MedBullets) benefit the most from collaborative deliberation, but biomedical literature interpretation (PubMedQA) and expert-level synthesis tasks (MedXpertQA-U) also show stable gains. This cross-domain robustness indicates that the benefit arises from the structure of the reasoning process, not from domain-specific heuristics or memorized knowledge.

Small, well-facilitated teams with evidence feedbacks optimize clinical reasoning

Team configuration and facilitation structure, rather than just panel size or model capacity, influence performance in multi-agent medical reasoning (Fig.3). Larger

teams boost accuracy in both retrieval and non-retrieval scenarios, but the improvement is more significant when search is unavailable. Without evidence access, additional clinicians offer diverse perspectives that partially offset missing information, resulting in larger accuracy gains. When search is available, each clinician already benefits from retrieved evidence, so adding more members yields smaller incremental improvements. Over discussion rounds, accuracy increases in both cases, with search-enabled teams consistently performing better in the end (Fig. 3b). Teams without retrieval see rapid early improvements but plateau at a lower level, whereas search-enabled teams keep refining their reasoning through multiple rounds. Across different coordination strategies, moderated meetings consistently outperform all other options (Fig. 3c). They score 3–7 percentage points higher than unmoderated chat, independent votes, or individual consultations. Moderated meetings also reach consensus faster, reducing disagreement entropy from 0.46 to 0.21, twice as quickly as unstructured discussion (Fig. 3e). The moderator achieves this by identifying conflicts, requesting targeted evidence to resolve disputes, and preventing premature closure when key questions remain unresolved. These patterns mirror clinical practice: multidisciplinary tumor boards succeed when participants have access to imaging and current literature, and effective facilitation ensures that all viewpoints are heard and that decisions are based on explicit evidence.

The system offers four discussion formats, varying from maximum to minimal coordination (Fig. 3d): moderated meetings with structured turns chaired by a moderator, unmoderated chat for free-form discussion, private individual consultations with an expert moderator, and independent voting where judgments are made in parallel without discussion. Among these, moderated meetings consistently outperform the others across all metrics. The results suggest a practical approach: use small panels of three experts with diverse specialties when retrieval is possible; conduct structured discussions with evidence verification; conclude after two rounds unless significant disagreement remains; and always cite sources. This approach aligns with the practices of high-performing clinical teams in tumor boards, ethics committees, and complex case conferences, suggesting MedAgents-X captures core principles of collaboration rather than task-specific strategies.

Retrieval quality stabilizes consensus through evidence reliability and coverage

To investigate how access to evidence influences group reasoning, we varied the retrieval methods while keeping the team structure unchanged (Fig. 4). Diagnosis quality was similar for corpus-only (31%) and web-only (30%) retrieval, but combining both sources (34%) led to more consistent consensus by grounding clinicians in trusted sources while still capturing recent information. When retrieval was completely disabled or replaced with random passages, consensus dropped significantly (25% and 23%), despite the same model capacity and discussion format. This difference indicates that retrieval is not just an information addition but a necessary

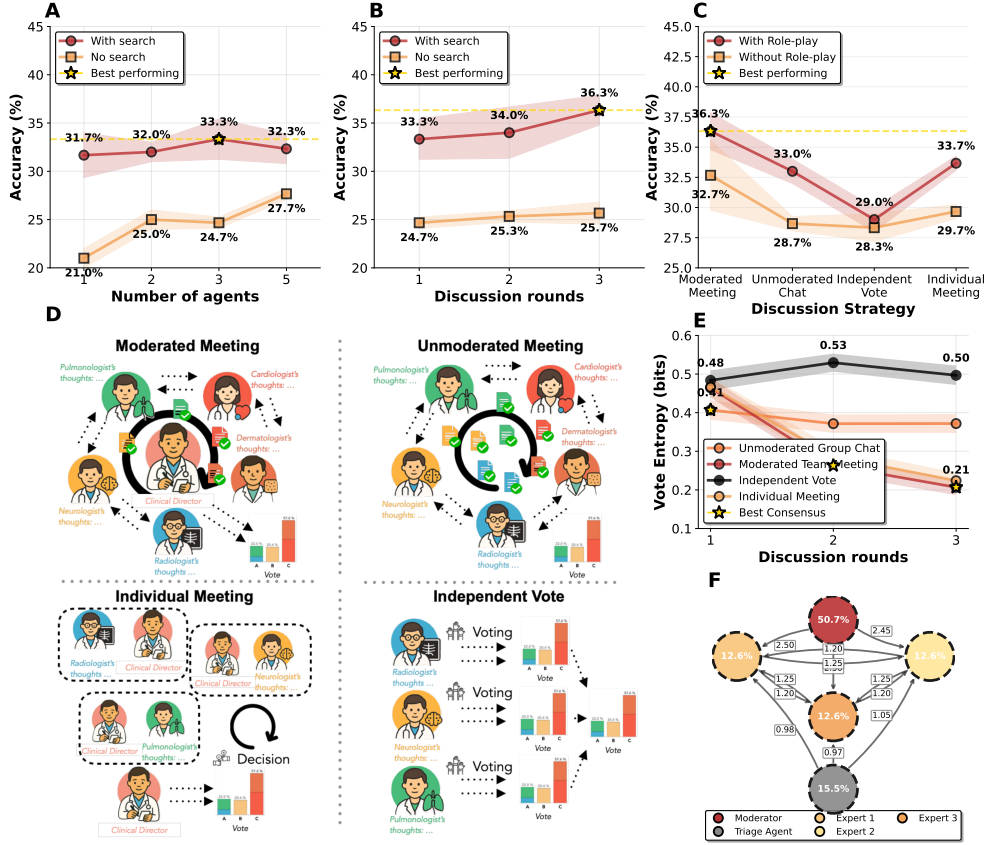


Fig. 3 Team composition, iteration, and facilitation design choices. (a) Larger teams improve performance only when retrieval is available: accuracy increases by ~ 4 –5 percentage points with additional agents under search. (b) Accuracy improves with additional discussion rounds in both settings, with the search-enabled condition achieving a higher final performance. Teams without retrieval show minimal gains. (c) Across coordination strategies, moderated meetings consistently outperform unmoderated chat, independent voting, and isolated one-on-one consultation, demonstrating that structured facilitation drives improvement. (d) Overview of the four deliberation paradigms. (e) Vote entropy falls by nearly $2\times$ under moderation, while remaining high in unmoderated settings. (f) Transition dynamics show that moderation corrects early disagreement rather than amplifying it.

condition for stable deliberation: without shared access to relevant evidence, agents keep reasoning but fail to reach agreement.

Two design choices further influence how retrieval stabilizes collaboration. Query rewriting broadens the search scope, enhancing recall by translating free-form clinical language into standardized medical terminology; document review improves precision by eliminating superficially plausible but irrelevant text. Removing either component decreases stability (31% without rewriting; 27% without review; Fig.4b), showing that the system needs both breadth and filtering to maintain epistemic alignment. Conversely, the complete setup with both features

maintained 33% accuracy at a moderate computational cost, demonstrating how rewriting and review complement each other by extending reach while reducing noise. Lastly, sharing search history among clinicians gave a small benefit (35%) but incurred more than five times the latency (Fig. 4c). Once a solid “evidence anchor” is established, added context rarely yields further benefits.

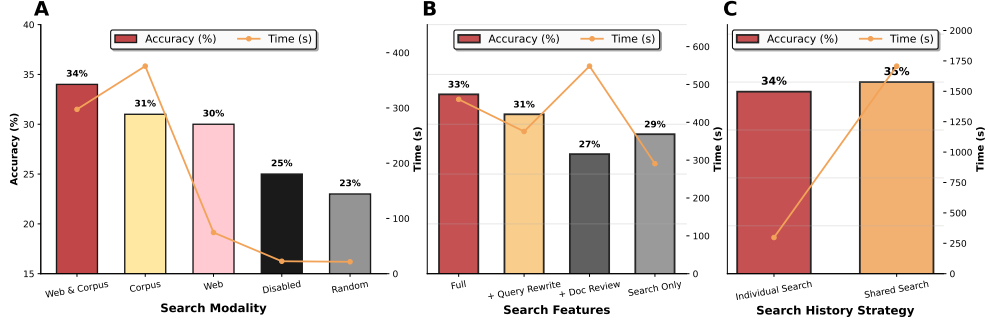


Fig. 4 Evidence retrieval strategies impact diagnostic accuracy and consultation efficiency. **a**, We compared the five search settings (i.e., Corpus, Web, Web & Corpus, Disabled, and Random), which reflect increasing breadth of accessible evidence. Combining corpus and web search yielded the most balanced accuracy and latency, whereas disabling retrieval or injecting random text sharply degraded performance. **b**, Removing either query rewriting (-2%) or document review (-6%) reduced accuracy and stability. Using document review alone, however, increased latency and reduced accuracy. **c**, Sharing search histories across clinicians slightly improved accuracy but extended consultation time significantly (more than 5 $\times$).

Adaptive triage scaffolds expertise through role-aligned information seeking

To explore why specialization enhances group reasoning, we examined the triage component that selects a small, role-diverse panel and provides each clinician with a brief specialty profile. This profile guides their evidence search and interpretation (Fig. 5a). This step mirrors the initial phase of real clinical consultations—determining relevant expertise and deciding the depth of discussion. Even this simple routing significantly decreases discussion drift by giving agents a stable interpretive framework before collaboration starts.

Role specialization proved to be more than just cosmetic; it was mechanistically significant: panels with realistic specialty profiles outperformed generic “medical expert” agents by 3.6 percentage points (Fig. 5b), even though they had no extra training or domain-specific parameters. The improvement came not from deeper knowledge, but from structured information-seeking: specialists used narrower queries, filtered irrelevant information more effectively, and avoided pattern-matching shortcuts that generic agents often relied on.

In various scenarios, triage naturally recruits specialties relevant to the specific problem. For instance, research-oriented questions like PubMedQA typically involve literature specialists such as clinical researchers. Conversely, diagnostic

questions, including MedBullets and MedQA, focus on primary care and emergency medicine, involving emergency and internal medicine professionals (see Fig. 5c). This spontaneous grouping, achieved through a simple matching prompt without explicit domain labels, indicates that question structure contains implicit cues about the specialty. Lightweight triage can reliably interpret these cues, similar to how human clinicians deduce consultation needs solely from case presentation.

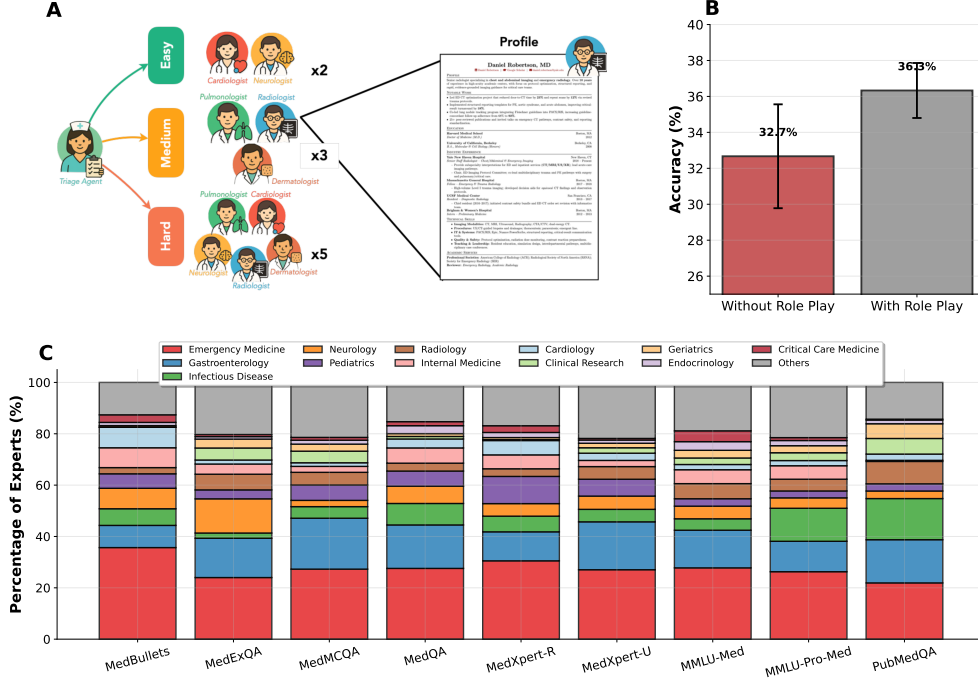


Fig. 5 Triage routes expertise and role-play sharpens retrieval. **a**, Triage workflow: estimate difficulty, assemble a small role-diverse panel, and generate specialty profiles that steer queries and evidence review. **b**, Role specialization improves accuracy by roughly 3–4 percentage points over generic agents at comparable latency and cost. **c**, Triage aligns specialties to dataset domains (e.g., Clinical Research 35.5% in PubMedQA, Emergency Medicine 19.0% in MedBullets, Internal Medicine 17.2% in MedQA), which correlates with the domain of the question.

Error Analysis and Human Evaluation

To examine how multi-agent deliberation fails and where improvement is most needed, we conducted a structured human evaluation of representative error trajectories and collected evidence (Fig. 6). Details of the annotation schema and error categories are provided in Table 1. Here, we focus on the emergent patterns and their implications for system design.

Across benchmarks, failures primarily arose from how evidence was interpreted and shared during discussion. As shown in Fig. 6a–b, *misled discussion* was the

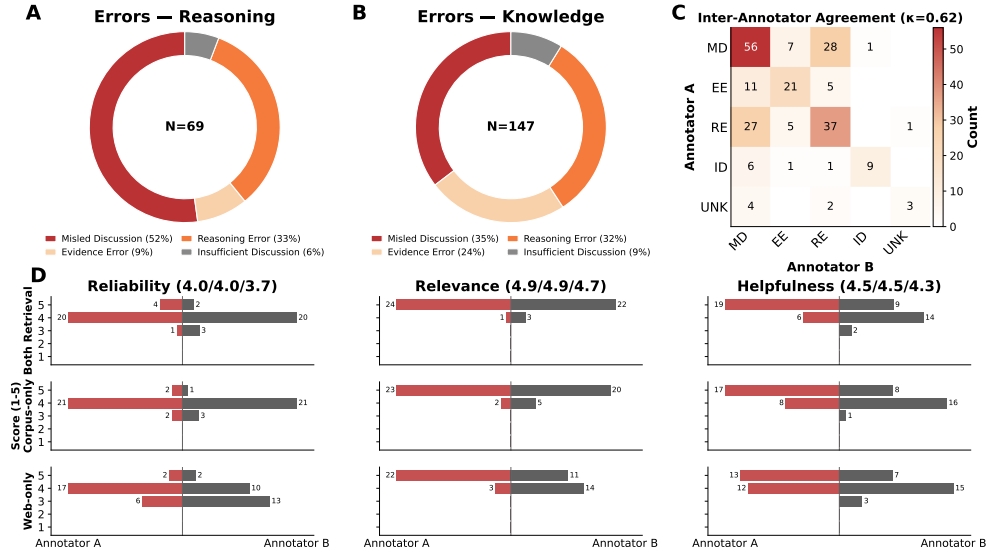


Fig. 6 Human evaluation of error modes and evidence quality. **a–b**, Across both the reasoning and knowledge views, misled discussion is the predominant failure mode, followed by reasoning errors; evidence errors are less common, and insufficient discussion is rare. **c**, Annotators show moderate agreement on the four labels (Cohen’s $\kappa = 0.62$), with most disagreements along the boundary between evidence quality and interpretation. **d**, Evidence quality improves with hybrid (web+corpus) and corpus-only retrieval compared with web-only, yielding higher reliability and helpfulness, and the highest relevance when combining sources.

predominant failure mode, often initiated by a plausible yet incorrect hypothesis that rapidly gained consensus and proved resistant to correction. *Reasoning errors* typically reflect accurate retrieval followed by misinterpretation, indicating that factual grounding alone does not ensure valid inference. In contrast, *evidence errors* and *insufficient discussion* were comparatively secondary, occurring when agents prematurely filtered decisive passages or closed debates before uncertainty was resolved. Despite moderate agreement among annotators ($\kappa = 0.62$, Fig. 6c), the variation largely reflects the complexity and inherent ambiguity of clinical reasoning trajectories rather than inconsistencies in labeling.

Finally, as shown in Fig. 6d, retrieval conditions greatly affected consensus stability: corpus-only and hybrid (web+corpus) retrieval yielded more reliable and relevant evidence, while web-only sources often introduced contextually suitable but methodologically weak material. Effective collaboration, therefore, depends on explicit mechanisms that encourage re-examination and balanced judgment, for example, revisiting evidence when opinions diverge and ensuring that differing perspectives are voiced before closure. Overall, these findings indicate that reliable collective reasoning arises not from greater information volume but from access to trustworthy evidence and well-structured facilitation that sustain critical evaluation throughout discussion.

Case study: Moderator-driven arbitration under conflicting evidence in atypical chest pain

This case highlights how moderation can mediate between conflicting yet individually valid reasoning paths. A 58-year-old woman presented with intermittent retrosternal discomfort radiating to the left arm and mildly elevated high-sensitivity troponin (0.045 ng/mL, reference <0.04), raising concern for unstable angina versus non-cardiac chest pain. In Round 1, the panel diverged: the cardiology expert recommended dual antiplatelet therapy and hospital admission per the 2020 ESC guidelines for non-ST-elevation acute coronary syndromes, while the internal medicine expert favored outpatient follow-up, citing a low pre-test probability and anxiety-related symptoms. The emergency medicine expert suggested serial troponin testing before deciding on therapy. Despite all citing authoritative evidence, their conclusions differed due to subtle differences in the timing and risk thresholds of the sources they referenced.

Instead of voting, the moderator started an evidence arbitration process, creating targeted search questions around three main uncertainties: (i) managing borderline troponin elevation, (ii) using risk scores in low-to-moderate chest pain, and (iii) diagnostic value of early noninvasive imaging. The new sources revealed a key nuance: recent meta-analyses and 2023 AHA/ACC guidance advised against immediate dual therapy for low-risk troponin rises and supported observation with serial testing.

In Round 2, the panel aligned around this combined evidence, agreeing on serial troponin testing and stress imaging as the next step. The cardiology expert changed their view, recognizing the risk of overtreatment in the initial guidelines. The correction was not due to extended debate but resulted from the moderator's focused organization of conflicting information. This case shows how MedAgents-X transcends simple opinion collection: moderation functions as an epistemic regulator that reshapes the shared evidence landscape, allowing rational consensus even when initial sources seem authoritative but are inconsistent.

Complete Trajectory: Atypical Chest Pain – Moderator-driven Arbitration under Conflicting Evidence

Question: A 58-year-old woman presents with intermittent retrosternal discomfort radiating to the left arm and mild dyspnea. Blood pressure 128/74 mmHg, HR 92 bpm, SpO₂ 98%. ECG: normal sinus rhythm. High-sensitivity troponin: 0.045 ng/mL (reference <0.04). Past history: gastroesophageal reflux disease and anxiety disorder. What is the most appropriate next step in management?

Options: A: Dual antiplatelet therapy and admission; B: Serial troponin testing and stress imaging; C: Proton pump inhibitor trial; D: Reassurance and discharge; E: Cardiac catheterization.

Ground truth vs. final. Ground truth: B (Serial troponin testing and stress imaging). Final consensus: B (Serial troponin testing and stress imaging).

Triage. Difficulty: high (ambiguous biomarkers, guideline inconsistency). Meeting: Moderated multi-specialty discussion. Panel size: 3. Rounds: 2.

Panel. Dr. Laura Chen (Cardiology, $w=0.4$); Dr. Anil Patel (Internal Medicine, $w=0.3$); Dr. Michael Torres (Emergency Medicine, $w=0.3$).

Moderator agenda. (1) Assess discordant guideline recommendations for borderline troponin elevation; (2) clarify pre-test probability thresholds; (3) identify consensus criteria for low-intermediate chest pain management.

Round 1 agenda. Compare recommendations and evidence bases for acute chest pain evaluation under conflicting guidelines.

- **Dr. Laura Chen (Cardiology)** — vote: A (high). Evidence (highlights): 2020 ESC guideline on NSTEMI-ACS recommends early dual antiplatelet therapy and admission when troponin exceeds assay cutoff. Rationale: Troponin elevation, even if mild, meets threshold for unstable angina; err on side of treatment.
- **Dr. Anil Patel (Internal Medicine)** — vote: D (medium). Evidence (highlights): 2017 ACP chest pain evaluation summary emphasizes frequent non-cardiac etiologies and recommends outpatient follow-up in low-risk cases. Rationale: Symptoms intermittent, no ischemic ECG change, likely functional or reflux-related.
- **Dr. Michael Torres (Emergency Medicine)** — vote: B (medium). Evidence (highlights): 2019 AHA scientific statement suggests serial troponin and imaging before initiating therapy in low-intermediate presentations. Rationale: Observe trend before committing to therapy.

Moderator action: Pause and diagnose source of disagreement. Identifies that agents use temporally inconsistent guidelines (ESC 2020 vs AHA 2019 vs ACP 2017). Initiates targeted retrieval to arbitrate evidence conflict on: (i) updated 2023 AHA/ACC chest pain guidance; (ii) risk score calibration (HEART, EDACS) in low-intermediate cases; (iii) meta-analyses comparing outcomes of early dual therapy vs. observation.

Round 2 agenda. Synthesize retrieved evidence and reassess management strategy.

- **Dr. Laura Chen (Cardiology)** — vote: B (high). Evidence (highlights): 2023 AHA/ACC guideline: for borderline troponin elevations (<0.05 ng/mL) with normal ECG, recommend risk-stratified testing rather than automatic dual therapy; overtreatment linked to bleeding risk. Rationale revised accordingly.
- **Dr. Anil Patel (Internal Medicine)** — vote: B (high). Evidence (highlights): HEART score ≤ 3 indicates $<2\%$ 30-day MACE risk; observation appropriate. Rationale: Evidence synthesis supports moderate monitoring, not discharge.
- **Dr. Michael Torres (Emergency Medicine)** — vote: B (high). Evidence (highlights): Updated meta-analysis (2022): observation with serial troponin yields equivalent outcomes to early dual therapy in low-intermediate risk patients. Rationale unchanged.

Moderator action: Conclude evidence arbitration. Consensus stabilized (B). Logs retrieved sources and guideline discrepancies to the knowledge base.

Consensus and outcome. Final: B (correct). *Notes.* The panel initially diverged due to temporally inconsistent guideline retrievals. Moderator-led arbitration resolved the

conflict by aligning evidence recency and relevance, shifting the decision from premature dual therapy to risk-stratified testing. Demonstrates MedAgents-X’s ability to reconcile authoritative but conflicting knowledge sources.

Audit trail elements. All queries, retrieved passages, and citations logged; excerpt includes representative retrieval summaries and rationale revisions.

Discussion

Facilitated collaboration produces stronger consensus. Our results show that evidence-grounded multi-agent collaboration can meaningfully improve the quality and defensibility of medical reasoning. Small, well-facilitated teams consistently outperform unmoderated or independent setups and reach consensus more efficiently when a moderator directs attention to key disagreements and evidence gaps. These patterns echo clinical experience: structured and chaired discussions, such as multidisciplinary tumor boards, tend to produce more transparent and accountable decisions than ad hoc deliberations. Earlier frameworks focused on demonstrating that multiple agents can reason together. In contrast, MedAgents-X reframes collaboration itself as a process that is structured, governed by evidence, and accountable rather than merely conversational.

Finding the right balance: team size, retrieval, and facilitation. Three design choices appear critical for reliable clinical reasoning. First, facilitation is not mere overhead. Moderator-guided discussions measurably reduce disagreement entropy and raise accuracy, underscoring the importance of active coordination as a core mechanism. Second, performance plateaus with three experts and two discussion rounds, revealing a practical equilibrium between expertise diversity and coordination cost. Third, hybrid retrieval—combining query rewriting and document review—proves essential for stabilizing reasoning, mitigating hallucinations, and producing audit-ready citations.

Translational implications for clinical consultations. The system’s behavior offers practical lessons for real-world integration of AI decision support. Small, role-diverse teams with explicit coordination can expose disagreements early, request targeted evidence, and reach timely consensus. Structured retrieval pipelines with shared search histories and document review minimize redundancy and ensure citation precision. Even lightweight triage that aligns specialties with case domains can enhance both efficiency and clinical face validity. Collectively, these insights provide a pragmatic blueprint for embedding AI copilots within existing consultation workflows—augmenting, rather than replacing, expert judgment.

Limitations and open questions. Several limitations warrant consideration. Our evaluation focuses on multiple-choice reasoning tasks, which capture only part of the open-ended diagnostic and longitudinal reasoning required in practice. The system also inherits the constraints of its retrieval corpus and may reflect temporal gaps

or embedded biases in underlying data. In addition, each configuration was executed once per dataset due to the high computational cost of large-scale multi-agent inference, precluding formal significance testing. Nonetheless, consistent improvements across nine datasets and three backbone models suggest stable trends rather than stochastic variance. Future work will include multi-seed replications and bootstrap confidence intervals to further assess robustness. As larger foundation models are incorporated, careful calibration and safety guardrails will remain essential for high-stakes clinical use.

Near-term pathways forward. We identify two near-term opportunities for deployment. First, embedding MedAgents-X as a decision-support assistant in tumor boards and specialty consultations could surface targeted evidence while generating auditable rationales. Second, the entropy and convergence metrics developed here may serve as real-time indicators of discussion quality, flagging cases that merit expanded consultation or additional evidence gathering. Ultimately, MedAgents-X demonstrates that small, well-coordinated expert teams, grounded in verifiable evidence, can enhance both the reliability and accountability of AI-assisted medical reasoning—offering a realistic path toward transparent, defensible clinical decision support.

Methods

Problem Definition

We model medical question answering as a structured clinical consultation. We formalize multiple-choice medical QA as decision-making under explicit evidence with auditable traces. Given a clinical case c and candidate diagnoses $\mathcal{A} = \{a_1, \dots, a_n\}$, the system must identify the most appropriate answer

$$a^* = \arg \max_{a_i \in \mathcal{A}} \sigma(a_i \mid c, \mathcal{E}), \quad (1)$$

where $\mathcal{E}$ is the set of supporting evidence drawn from authoritative, up-to-date sources, and $\sigma(\cdot)$ represents the decision score computed from clinical evidence $\mathcal{E}$. The output includes the selected answer a^* , the cited evidence ϵ , and a complete audit trail $\mathcal{T}$ of the decision process, mirroring clinical requirements for decisions that are both accurate and defensible. This process is referred to as *consultation*.

To probe reasoning rather than recall, we evaluate on a deliberately challenging subset of medical questions—cases that remain unsolved by over half of current state-of-the-art models. This **hard subset** reflects complex, multi-step reasoning scenarios that stress-test the limits of current medical AI systems, emphasizing interpretability and collaboration over memorization.¹

¹Lower absolute accuracies across all models reflect this deliberate focus on diagnostically difficult questions that require reasoning under uncertainty.

MedAgents-X Architecture

MedAgents-X operationalizes the consultation paradigm through a multi-agent architecture designed to resemble a digital clinical team. The system comprises distinct agents, *moderator*, *clinicians*, and *triage*, working together to analyze cases, retrieve evidence, deliberate, and reach a transparent consensus.

Agents. Every agent in MedAgents-X is defined by its role, expertise profile, and consultation responsibilities. Profiles include credentials, background, and specialty focus, spanning 49 medical disciplines defined by the American Board of Medical Specialties.² Together, they form a virtual panel that mirrors multidisciplinary case discussions in clinical practice.

- **Moderator agent.** The moderator guides the consultation toward a defensible conclusion. It sets the agenda, clarifies the question, highlights disagreements, and requests additional evidence when gaps arise. The moderator synthesizes evolving viewpoints across rounds and determines when consensus is sufficient to conclude.
- **Clinician agents.** Each clinician agent represents a domain specialist (e.g., cardiology, infectious disease) responsible for reasoning within its field. Clinicians perform evidence-grounded analysis over a curated ~ 2.45 -million-segment medical corpus and the entire Internet. They refine their queries with specialty-aware rewriting, review retrieved passages, and construct short, cited rationales with calibrated confidence scores.
- **Triage agent.** The triage agent serves as the system’s coordinator. It evaluates case complexity, selects the appropriate mix of specialties, assigns relevance weights to each clinician, and determines how many rounds of discussion are warranted.

Evidence synthesis. Evidence synthesis is based on current, verifiable data. Each consultation combines specialty-based query rewriting [32, 33], dense retrieval with MedCPT encoders [34], and Milvus [35] HNSW indexing over a curated 2.45 million segment corpus. Candidate passages are reranked, filtered for specificity, and supplemented by web retrieval as needed. A similarity cache reuses results for queries that closely match ($\tau \geq 0.85$), reducing redundancy. Clinicians record concise, cited rationales with confidence scores on an audit trail. This asynchronous process runs across specialists in up to three rounds, mimicking real consultations.

Clinicians interpret each case from their specialty, identify discriminative findings (*reasoning*); refine queries with a medical rewriter that expands abbreviations and aligns terminology to domain standards (*query rewriting*); retrieve and rerank relevant passages, filter low-specificity items (*evidence retrieval*); and synthesize evidence into a concise, cited rationale with a confidence score (*synthesis*).

Medical Corpus. Following Med-PRM [36], we employ the MedCPT bi-encoder and cross-encoder models [34] for dense retrieval and reranking. As in Med-PRM, we apply recursive text chunking to construct the retrieval corpus, extending it to

²<https://www.abms.org/abms-guide-to-medical-specialties/>

include five biomedical sources: Clinical Guidelines [37], StatPearls [16], Medical Textbooks [23], PubMed Abstracts [34], and the Rare Disease Corpus [38], resulting in context-preserving passages suitable for embedding and retrieval. For each query, 100 documents per corpus (400 total) are retrieved, with top 32 selected after reranking.

Clinical Diagnoses via MedAgents-X

We apply MedAgents-X to the problem of clinical diagnosis. A case enters a consultation workflow that (i) assembles a small, role-diverse panel from 49 specialties, (ii) conducts MedCPT-based dense retrieval over a curated 2.45M-segment corpus with optional web retrieval, (iii) runs one to three discussion rounds determined by triage to resolve disagreements and request targeted follow-ups, and (iv) aggregates expert recommendations with citations into a final answer and auditable trace.

Triage. Triage serves as the gateway to every consultation, aligning the expertise mix and deliberation depth with case complexity. Each case is first evaluated across five criteria—knowledge depth, reasoning complexity, clinical ambiguity, terminological specificity, and rarity. This assessment produces a difficulty label $\gamma \in \{\text{easy}, \text{medium}, \text{hard}\}$, accompanied by a short rationale summarizing why the case falls into that tier.

Given γ , the triage agent automatically assembles a small, role-diverse panel drawn from 49 medical specialties. Panel size scales with complexity: two specialists for easy cases, three for medium, and five for hard. Specialties are chosen to maximize coverage of relevant organ systems and viewpoints—for example, a complex cardiovascular infection may involve cardiology, infectious disease, and nephrology experts.

Each clinician receives a relevance weight w_i (normalized so that $\sum_i w_i = 1$), reflecting how central their specialty is to the case. These weights persist through the consultation, influencing both evidence retrieval and the final consensus vote. Finally, triage sets the deliberation budget—one, two, or three rounds depending on difficulty—defining the structure of the discussion before the moderator convenes the panel.

Multi-Agent Discussion. Consultations are organized as structured “discussions” with an *agenda*, optional *questions* and *context*, and a triage-defined number of rounds ($N=1-3$). Each format differs in interaction style and degree of coordination.

- **Moderated meeting.** All clinicians and the moderator meet in a shared thread. The moderator opens the agenda, assigns focus areas, and manages turns. Clinicians present cited evidence, debate discrepancies, and revise votes each round. The moderator highlights conflicts, requests follow-up retrievals, and summarizes emerging consensus. The meeting ends when the weighted top choice stabilizes or all disagreements are resolved.

- **Unmoderated meeting.** Clinicians converse freely without coordination, responding to one another’s messages and attached evidence. After the fixed rounds, the leading answer by weighted vote share becomes the consensus.
- **Individual meeting.** Each clinician meets privately with the moderator, who tailors prompts to their specialty. The moderator integrates all cited recommendations into a single relevance- and confidence-weighted decision.
- **Independent vote.** Each clinician submits one cited rationale and confidence score; the system aggregates a single weighted vote in one pass—fast but least interactive.

All meeting types record agendas, queries, retrieved passages, rationales, votes, and the final stopping rationale to form the audit trail $\mathcal{T}$.

Voting and consensus. Each discussion round functions like a mini case conference. Every clinician i contributes both a *vote* $a_i^{(r)}$, and a *confidence score* $\rho_i^{(r)} \in [0, 1]$, weighted by their clinical relevance w_i . These inputs are combined into per-option support values

$$u_k^{(r)} = \sum_{i=1}^M w_i \rho_i^{(r)} \mathbb{1}[a_i^{(r)} = a_k], \quad (2)$$

which are then transformed into a temperature-controlled softmax consensus distribution [39]

$$p_k^{(r)} = \frac{\exp(u_k^{(r)} / \tau)}{\sum_{j=1}^K \exp(u_j^{(r)} / \tau)}, \quad \hat{k}^{(r)} = \arg \max_k p_k^{(r)}. \quad (3)$$

Intuitively, this process resembles a moderated panel vote: high-confidence clinicians exert stronger influence, while the temperature parameter τ tunes how easily minority opinions can shift the outcome. After each round, the moderator examines the gap between the top two probabilities—our operational measure of *decisiveness*. If the leading choice remains stable beyond a preset threshold (or the allotted rounds are complete), the consultation concludes. The final output includes the consensus answer $\hat{k}^{(r)}$, aggregated confidence, and the full citation trail supporting the decision.

Datasets and Evaluation Protocol

We evaluate our approach across eight established medical QA datasets [23–30], spanning clinical knowledge, case-based reasoning, and biomedical literature interpretation. To address ceiling effects observed in standard evaluations, we construct MedAgentsBench [31], a curated subset of 862 challenging questions that require complex multi-step reasoning. All methods follow standardized evaluation protocols with three independent random seeds, identical hardware (NVIDIA A100, 40GB RAM), temperature 0.1, and maximum output length 4096 tokens. We report

accuracy as the primary metric, with inference time and token-based cost as secondary metrics.

- **MedQA** [23]: Professional medical board exam questions from USMLE and similar licensing examinations, testing comprehensive medical knowledge including diagnosis, treatment, and clinical reasoning across diverse medical domains.
- **PubMedQA** [24]: Biomedical research questions derived from PubMed abstracts requiring reasoning over quantitative findings and statistical evidence rather than simple medical fact recall for yes/no/maybe conclusions.
- **MedMCQA** [25]: Real-world medical entrance exam questions from AIIMS and NEET PG covering 2,400+ healthcare topics and 21 medical subjects, emphasizing practical application over memorization.
- **MedBullets** [26]: USMLE Step 2/3 style clinical vignettes from Medbullets platform, focusing on challenging realistic scenarios requiring integration of medical knowledge with clinical reasoning.
- **MedExQA** [29]: Multi-specialty questions spanning five underrepresented domains (Audiology, Nutrition, Occupational Therapy, Physical Therapy, Speech-Language Pathology) with multiple expert explanations for comprehensive evaluation.
- **MedXpertQA** [30]: Expert-level medical reasoning benchmark spanning 17 specialties and 11 body systems, divided into understanding (MedXpertQA-U) and reasoning (MedXpertQA-R) subsets for comprehensive clinical decision-making assessment.
- **MMLU (Medical)** [40]: Medical knowledge questions covering clinical knowledge, anatomy, college medicine, medical genetics, and professional medicine across elementary to advanced professional difficulty levels.
- **MMLU-Pro (Medical)** [28]: Enhanced MMLU version eliminating trivial questions while incorporating challenging, reasoning-focused alternatives with expanded choice sets to address ceiling effects in advanced models.

As in [31], we build a discriminative benchmark by targeting items where leading models still fail. Each question was evaluated across 10 systems (GPT-4o, GPT-4o-mini, Claude-3.5-Sonnet, Claude-3.5-Haiku, o1-mini, o3-mini, DeepSeek-V3, DeepSeek-R1, Llama-3.3-70B, QwQ-32B), and we retained only those answered correctly by $< 50\%$ of the models, filtering out rote facts and enriching for multistep clinical reasoning. The final set is balanced: 100 questions each from MedQA, PubMedQA, MedMCQA, MMLU-Pro, MedExQA, and both MedXpertQA subsets, plus 89 from MedBullets and 73 from MMLU. Then four medical professionals reviewed all items to confirm clinical relevance and that success requires genuine reasoning rather than recall. MELD-based contamination analysis reveals that OpenAI models consistently exhibit lower memorization across datasets; accordingly, we adopt them as our base models.

For ablation studies, each evaluation was conducted once per dataset (MedQA) and backbone (gpt-4o-mini) due to high computational costs. Although this prevents formal significance testing, consistent improvements across nine datasets and three model families suggest stable trends rather than stochastic variability.

Baselines

We compare against zero-shot, few-shot, chain-of-thought, self-consistent CoT, MedRAG, Self-Refine, MedPrompt, MultiPersona, MedAgents, MDAgents, AFlow, and SPO, following published or open-source configurations with GPT-4o(-mini) as the executor model and structured JSON output parsing for consistent answer extraction.

- **Zero-shot**: The model is guided solely by task instructions and produces an answer without any exemplars.
- **Few-shot** [41]: The model performs in-context learning from a small set of demonstrations included in the prompt; we provide five exemplars sampled from the training set.
- **Chain-of-Thought (CoT)** [42]: The prompt asks the model to externalize intermediate, step-by-step reasoning before committing to a final answer, which improves arithmetic and symbolic reasoning.
- **Self-Consistent CoT (CoT-SC)** [43]: Instead of relying on one chain, the model samples multiple independent reasoning trajectories and aggregates the answers by a majority vote, thereby marginalizing over spurious chains. We provide five independent paths for ensemble aggregation.
- **MedRAG** [16]: The model performs retrieval-augmented generation over a medical corpus (PubMed, StatPearls, textbooks, and Wikipedia), using MedCPT as the dense retriever; we prepend $k=32$ passages to the prompt and retain CoT prompting for transparency.
- **Self-Refine** [44]: The model iteratively critiques and revises its own draft, alternating between targeted feedback and editing; we run up to three refinement rounds with early stopping (if confidence ≥ 0.95) on self-rated confidence.
- **MedPrompt** [7]: This method composes general prompting strategies, such as few-shot examples, chain of thought, assembly/shuffling, and prompt self-design, to reduce positional bias and stabilize medical QA. Configured with $k=3$ retrieved exemplars, three generations, and three voting rounds.
- **MultiPersona** [45]: A single model instantiates multiple complementary personas (e.g., innovator, analyst, clinician) that debate for several turns and then synthesize a final solution under the guidance of a senior arbiter.
- **MedAgents** [21]: A training-free, multi-agent framework coordinates role-specialized experts that independently analyze a case, discuss, and converge to a consensus decision via a synthesize-verify protocol. We employ five agents, allowing up to three voting attempts and revisions until unanimity is reached.

- **MDAgents** [22]: An adaptive controller selects solo versus group collaboration based on assessed case complexity, optionally incorporating an external knowledge source and a moderator to arbitrate disagreements. We employ 2–3 agents per tier, one to three consultations, and expertise-weighted voting.
- **AFlow** [46]: Agentic workflows are represented as executable code graphs composed from reusable operators (e.g., Generate, Revise, Ensemble, Test); a Monte-Carlo Tree Search explores this space and early-stops on convergence. We provide fixed executor models and twenty search rounds with early stopping.
- **SPO** [47]: Prompt candidates are iteratively proposed, executed, and pairwise-evaluated by an LLM judge to enable label-free prompt search. We set the number of iterations to ten, with three sampled queries per round, and GPT-4o-mini as both the executor and evaluator.

Human Evaluation

Two complementary evaluations were performed to assess MedAgents-X reasoning and retrieval capabilities. Here, annotators followed standardized guidelines: (1) read the full transcript, (2) verify answer correctness, (3) trace evidence flow, (4) assign one error label in a structured spreadsheet.

Failure Mode. We performed a failure analysis on 225 representative consultations (25 per dataset, o3-mini backbone) to characterize reasoning failures in MedAgents-X. Each transcript, including agent thoughts, cited evidence, and final votes, was independently reviewed by two licensed physicians with MD degrees who are in an internal medicine residency program at a North American medical school, using a predefined rubric. Annotators identified the dominant error type, contributing factors, and mitigation notes. Disagreements were adjudicated by an MD-PhD physician with clinical training. Inter-annotator agreement was substantial ($\kappa = 0.62$). Table 1 summarizes the five non-exclusive error categories used during annotation.

Table 1 Taxonomy of reasoning errors used for manual annotation.

Failure mode	Definition
Misled Discussion	The group judgment was biased towards incorrect claims.
Evidence Error	Retrieved sources were irrelevant or contradictory to the case.
Reasoning Error	Correct evidence was retrieved, but misinterpreted or ignored.
Insufficient Discussion	Discussion ended before key disagreements were resolved.

Evidence Quality. We also performed an evidence quality assessment on 75 representative consultations (25 per case, gpt-4o-mini backbone). The quality of the

recovered passages was manually assessed using three complementary criteria, *Reliability*, *Relevance*, and *Helpfulness*. Two practicing clinician annotators independently rated each passage on a 5-point scale (1. very low, 5. very high) according to the definitions in Table 2.

Table 2 Criteria and 5-point scale for manual annotation of retrieval evidence quality.

Criterion	Annotation Guideline
Reliability	1. Non-authoritative or unverifiable material. 2. Dated or low-impact source. 3. Educational or secondary summary. 4. Specialty textbook or systematic review. 5. Authoritative guideline or peer-reviewed clinical reference.
Relevance	1. Tangential or off-topic. 2. Weakly related or missing key discriminative features. 3. Partially aligned or generic within the same disease class. 4. Closely related to minor contextual gaps. 5. Direct match to disease, presentation, and patient context.
Helpfulness	1. Non-informative or misleading content. 2. Minimal diagnostic impact or redundant information. 3. Provides general background without resolving ambiguity. 4. Contributes to diagnostic reasoning or next-step decision. 5. Directly supports or refutes a key differential diagnosis.

Implementation Details. We implement MedAgents-X using the OpenAI Agents SDK with asynchronous execution for parallel expert processing. Vector search uses Milvus 2.5.6 with HNSW indexes and MedCPT embeddings precomputed on NVIDIA A100 GPUs. Agents communicate via structured JSON validated with Pydantic, and a usage tracker records input/output tokens and cache hits. We employ exponential backoff (max five attempts) to handle transient API failures.

Supplementary information. If your article has accompanying supplementary file/s please state so here.

Authors reporting data from electrophoretic gels and blots should supply the full unprocessed scans for key as part of their Supplementary information. This may be requested by the editorial team/s if it is missing.

Please refer to Journal-level guidance for any specific requirements.

Data Availability. The MedAgentsBench dataset [31] is publicly available at <https://huggingface.co/datasets/super-dainiu/medagents-benchmark>.

Code Availability. Source code, configurations, evaluation scripts, and baselines are available at <https://github.com/gersteinlab/MedAgents-2> and <https://github.com/gersteinlab/medagents-benchmark>. Implementations use the OpenAI

Agents SDK (<https://github.com/openai/openai-agents-python>), Milvus [35], and MedCPT [34]. The curated corpus includes clinical practice guidelines, StatPearls, textbooks, and RECOP recommendations. Full setup and reproducibility materials are provided in the repositories.

Competing Interests. The authors declare no competing interests.

Ethics Statement. This work evaluates de-identified public benchmarks and does not involve patient data. MedAgents-X is intended as decision support, not a substitute for clinical judgment. Outputs may inherit biases from models and data; all recommendations should be verified by qualified clinicians before any clinical use.

References

- [1] Rodziewicz, T. L. & Hipkind, J. E. Medical error prevention. *StatPearls. Treasure Island (FL): StatPearls Publishing* (2020).
- [2] Bates, D. W. *et al.* Ten commandments for effective clinical decision support: making the practice of evidence-based medicine a reality. *Journal of the American Medical Informatics Association* **10**, 523–530 (2003).
- [3] Sutton, R. T. *et al.* An overview of clinical decision support systems: benefits, risks, and strategies for success. *NPJ digital medicine* **3**, 17 (2020).
- [4] Thirunavukarasu, A. J. *et al.* Large language models in medicine. *Nature medicine* **29**, 1930–1940 (2023).
- [5] Wang, Z. *et al.* Accelerating clinical evidence synthesis with large language models. *arXiv preprint arXiv:2406.17755* (2024).
- [6] Moor, M. *et al.* Foundation models for generalist medical artificial intelligence. *Nature* **616**, 259–265 (2023).
- [7] Nori, H. *et al.* Can generalist foundation models outcompete special-purpose tuning? case study in medicine. *arXiv preprint arXiv:2311.16452* (2023).
- [8] Singhal, K. *et al.* Large language models encode clinical knowledge. *Nature* **620**, 172–180 (2023).
- [9] Singhal, K. *et al.* Towards expert-level medical question answering with large language models. *arXiv preprint arXiv:2305.09617* (2023).
- [10] Li, J. *et al.* Agent hospital: A simulacrum of hospital with evolvable medical agents. *arXiv preprint arXiv:2405.02957* (2024).
- [11] Xie, Y. *et al.* A preliminary study of o1 in medicine: Are we closer to an ai doctor? *arXiv preprint arXiv:2409.15277* (2024).

- [12] Jin, Q. *et al.* Matching patients to clinical trials with large language models. *Nature Communications* **15**, 9074 (2024).
- [13] Pal, A., Umapathi, L. K. & Sankarasubbu, M. *Med-halt: Medical domain hallucination test for large language models*, 314–334 (2023).
- [14] Gilbert, S., Kather, J. N. & Hogan, A. Augmented non-hallucinating large language models as medical information curators. *NPJ Digital Medicine* **7**, 100 (2024).
- [15] Zakka, C. *et al.* Almanac—retrieval-augmented language models for clinical medicine. *NEJM AI* **1**, AIoa2300068 (2024).
- [16] Xiong, G., Jin, Q., Lu, Z. & Zhang, A. Benchmarking retrieval-augmented generation for medicine. *arXiv preprint arXiv:2402.13178* (2024).
- [17] Wu, J. *et al.* Medical graph rag: Towards safe medical large language model via graph retrieval-augmented generation. *arXiv preprint arXiv:2408.04187* (2024).
- [18] Jeong, M., Sohn, J., Sung, M. & Kang, J. Improving medical reasoning through retrieval and self-reflection with retrieval-augmented large language models. *Bioinformatics* **40**, i119–i129 (2024).
- [19] Xiong, G. *et al.* *Improving retrieval-augmented generation in medicine with iterative follow-up questions*, 199–214 (World Scientific, 2024).
- [20] Chen, X. *et al.* Enhancing diagnostic capability with multi-agents conversational large language models. *NPJ digital medicine* **8**, 159 (2025).
- [21] Tang, X. *et al.* MedAgents: Large language models as collaborators for zero-shot medical reasoning. *arXiv preprint arXiv:2311.10537* (2023).
- [22] Kim, Y. *et al.* Adaptive collaboration strategy for llms in medical decision making. *arXiv preprint arXiv:2404.15155* (2024).
- [23] Jin, D. *et al.* What disease does this patient have? A large-scale open domain question answering dataset from medical exams. *Applied Sciences* **11**, 6421 (2021).
- [24] Jin, Q., Dhingra, B., Liu, Z., Cohen, W. & Lu, X. *PubMedQA: A dataset for biomedical research question answering*, 2567–2577 (2019).
- [25] Pal, A., Umapathi, L. K. & Sankarasubbu, M. *MedMCQA: A large-scale multi-subject multi-choice dataset for medical domain question answering*, 248–260 (PMLR, 2022).

- [26] Chen, H., Fang, Z., Singla, Y. & Dredze, M. Benchmarking large language models on answering and explaining challenging medical questions. *arXiv preprint arXiv:2402.18060* (2024).
- [27] Hendrycks, D. *et al.* Measuring massive multitask language understanding. *arXiv preprint arXiv:2009.03300* (2020).
- [28] Wang, Y. *et al.* MMLU-Pro: A more robust and challenging multi-task language understanding benchmark. *arXiv preprint arXiv:2406.01574* (2024).
- [29] Kim, Y., Wu, J., Abdulle, Y. & Wu, H. MedExQA: Medical question answering benchmark with multiple explanations. *arXiv preprint arXiv:2406.06331* (2024).
- [30] Zuo, Y. *et al.* MedXpertQA: Benchmarking expert-level medical reasoning and understanding. *arXiv preprint arXiv:2501.18362* (2025).
- [31] Tang, X. *et al.* Medagentsbench: Benchmarking thinking models and agent frameworks for complex medical reasoning. *arXiv preprint arXiv:2503.07459* (2025).
- [32] Wang, L., Yang, N. & Wei, F. Query2doc: Query expansion with large language models. *arXiv preprint arXiv:2303.07678* (2023).
- [33] Ma, X., Gong, Y., He, P., Zhao, H. & Duan, N. *Query rewriting in retrieval-augmented large language models*, 5303–5315 (2023).
- [34] Jin, Q. *et al.* Medcpt: Contrastive pre-trained transformers with large-scale pubmed search logs for zero-shot biomedical information retrieval. *Bioinformatics* **39**, btad651 (2023).
- [35] Wang, J. *et al.* Milvus: A purpose-built vector data management system, 2614–2627 (2021).
- [36] Yun, J. *et al.* Med-prm: Medical reasoning models with stepwise, guideline-verified process rewards. *arXiv preprint arXiv:2506.11474* (2025).
- [37] Chen, Z. *et al.* Meditron-70b: Scaling medical pretraining for large language models. *arXiv preprint arXiv:2311.16079* (2023).
- [38] Wang, G. *et al.* Assessing and enhancing large language models in rare disease question-answering. *arXiv preprint arXiv:2408.08422* (2024).
- [39] Taubenfeld, A. *et al.* Confidence improves self-consistency in llms. *arXiv preprint arXiv:2502.06233* (2025).

- [40] Hendrycks, D. *et al.* Measuring massive multitask language understanding. *arXiv preprint arXiv:2009.03300* (2020).
- [41] Brown, T. B. *et al.* Language models are few-shot learners. *arXiv preprint arXiv:2005.14165* (2020).
- [42] Wei, J. *et al.* Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems* **35**, 24824–24837 (2022).
- [43] Wang, X. *et al.* Self-consistency improves chain of thought reasoning in language models. *arXiv preprint arXiv:2203.11171* (2022).
- [44] Madaan, A. *et al.* Self-Refine: Iterative refinement with self-feedback. *Advances in Neural Information Processing Systems* **36** (2024).
- [45] Wang, Z. *et al.* Unleashing the emergent cognitive synergy in large language models: A task-solving agent through multi-persona self-collaboration, 257–279 (2024).
- [46] Zhang, J. *et al.* Aflow: Automating agentic workflow generation. *arXiv preprint arXiv:2410.10762* (2024).
- [47] Xiang, J. *et al.* Self-supervised prompt optimization. *arXiv preprint arXiv:2502.06855* (2025).

Appendix A Original Table and Results

Figure A1 shows the comprehensive performance comparison across all datasets and models. Table A1 presents detailed accuracy results with statistical significance testing.

Table A1 Main Results: Accuracy (%) comparison across medical QA datasets and base models. Best results are **bolded**, second best are underlined. Results show mean $\pm$ std over 3 runs.

Method	MedQA	MedMCQA	PubMedQA	MMLU-Med	MedBullets	MedExQA	MedXpert-R	MedXpert-U	MMLU-Pro	Average
GPT-4o										
Zero-shot	33.3 $\pm$ 1.5	23.7 $\pm$ 1.2	8.3 $\pm$ 1.2	25.6 $\pm$ 1.6	21.0 $\pm$ 1.7	15.7 $\pm$ 2.1	9.3 $\pm$ 2.1	8.3 $\pm$ 2.1	19.7 $\pm$ 1.5	18.3 $\pm$ 8.2
Few-shot	28.7 $\pm$ 0.6	31.3 $\pm$ 2.5	20.0 $\pm$ 3.0	26.0 $\pm$ 1.4	24.3 $\pm$ 1.3	24.3 $\pm$ 1.5	13.7 $\pm$ 1.5	10.7 $\pm$ 2.5	13.7 $\pm$ 4.5	21.4 $\pm$ 6.9
CoT	44.0 $\pm$ 4.4	29.0 $\pm$ 1.0	11.3 $\pm$ 3.2	31.0 $\pm$ 4.4	30.7 $\pm$ 2.3	21.0 $\pm$ 2.6	14.3 $\pm$ 2.1	14.0 $\pm$ 1.0	37.7 $\pm$ 2.3	25.9 $\pm$ 10.7
CoT-SC	42.3 $\pm$ 5.0	<u>32.7 $\pm$ 2.5</u>	7.3 $\pm$ 1.5	31.5 $\pm$ 1.4	31.1 $\pm$ 0.7	20.7 $\pm$ 2.3	14.0 $\pm$ 4.6	13.7 $\pm$ 1.5	<u>39.7 $\pm$ 4.2</u>	25.9 $\pm$ 11.7
Self-refine	<u>45.3 $\pm$ 3.8</u>	29.3 $\pm$ 4.0	15.0 $\pm$ 2.0	36.1 $\pm$ 2.1	33.3 $\pm$ 5.1	20.3 $\pm$ 1.5	16.0 $\pm$ 1.0	16.7 $\pm$ 2.1	33.3 $\pm$ 3.1	27.3 $\pm$ 10.1
MedPrompt	37.3 $\pm$ 1.5	28.0 $\pm$ 1.0	11.0 $\pm$ 3.0	32.4 $\pm$ 0.8	24.7 $\pm$ 3.0	21.0 $\pm$ 1.0	14.3 $\pm$ 1.5	13.3 $\pm$ 0.6	36.3 $\pm$ 3.5	24.3 $\pm$ 9.4
MedRAG	37.0 $\pm$ 1.7	27.3 $\pm$ 2.1	18.7 $\pm$ 1.5	35.6 $\pm$ 4.1	25.8 $\pm$ 1.2	16.0 $\pm$ 3.5	10.3 $\pm$ 0.6	13.0 $\pm$ 1.7	34.7 $\pm$ 2.5	24.3 $\pm$ 9.6
MultiPersona	45.0 $\pm$ 1.0	29.7 $\pm$ 4.2	17.3 $\pm$ 2.1	34.7 $\pm$ 2.9	34.5 $\pm$ 4.6	21.0 $\pm$ 2.0	16.7 $\pm$ 5.8	13.0 $\pm$ 2.6	39.3 $\pm$ 2.5	27.9 $\pm$ 10.7
MDAgents	37.3 $\pm$ 1.5	27.0 $\pm$ 1.0	20.7 $\pm$ 2.5	28.8 $\pm$ 3.6	19.9 $\pm$ 5.5	21.3 $\pm$ 1.2	10.7 $\pm$ 2.3	11.7 $\pm$ 1.5	<u>39.7 $\pm$ 1.5</u>	24.1 $\pm$ 9.5
AFlow	47.0 $\pm$ 1.0	31.3 $\pm$ 1.5	<u>23.3 $\pm$ 4.6</u>	<u>37.9 $\pm$ 3.5</u>	35.9 $\pm$ 2.0	23.0 $\pm$ 2.6	14.3 $\pm$ 2.3	<u>19.7 $\pm$ 1.5</u>	36.0 $\pm$ 3.6	<u>29.8 $\pm$ 9.8</u>
SPO	32.0 $\pm$ 1.7	30.0 $\pm$ 2.0	31.3 $\pm$ 1.5	34.7 $\pm$ 2.1	28.8 $\pm$ 0.6	21.0 $\pm$ 2.0	<u>17.0 $\pm$ 2.0</u>	17.3 $\pm$ 1.5	36.7 $\pm$ 0.6	27.6 $\pm$ 7.0
MedAgents	43.0 $\pm$ 1.0	30.3 $\pm$ 0.6	17.7 $\pm$ 2.5	29.2 $\pm$ 4.8	20.3 $\pm$ 1.5	20.3 $\pm$ 1.5	13.0 $\pm$ 1.7	12.3 $\pm$ 2.1	37.7 $\pm$ 2.5	25.9 $\pm$ 10.1
MedAgents-X	44.0 $\pm$ 1.0	33.0 $\pm$ 1.0	19.3 $\pm$ 1.5	47.9 $\pm$ 1.4	<u>35.4 $\pm$ 1.7</u>	<u>24.0 $\pm$ 1.0</u>	22.3 $\pm$ 3.5	30.4 $\pm$ 2.7	41.0 $\pm$ 1.0	33.0 $\pm$ 9.4
GPT-4o-mini										
Zero-shot	21.3 $\pm$ 2.1	17.3 $\pm$ 0.6	11.0 $\pm$ 1.0	14.6 $\pm$ 2.1	9.4 $\pm$ 0.6	5.7 $\pm$ 1.5	5.7 $\pm$ 0.6	5.3 $\pm$ 0.6	10.7 $\pm$ 1.5	11.2 $\pm$ 5.3
Few-shot	30.3 $\pm$ 2.5	27.0 $\pm$ 3.5	22.0 $\pm$ 4.0	25.6 $\pm$ 2.9	22.5 $\pm$ 1.2	26.0 $\pm$ 1.0	<u>14.7 $\pm$ 1.5</u>	8.3 $\pm$ 0.6	9.7 $\pm$ 0.6	<u>20.7 $\pm$ 7.5</u>
CoT	24.7 $\pm$ 3.2	23.3 $\pm$ 2.5	14.3 $\pm$ 1.5	26.1 $\pm$ 2.4	17.6 $\pm$ 0.6	15.7 $\pm$ 1.5	7.3 $\pm$ 2.3	12.3 $\pm$ 2.5	<u>35.7 $\pm$ 2.1</u>	19.7 $\pm$ 8.1
CoT-SC	21.7 $\pm$ 2.1	22.3 $\pm$ 2.1	13.3 $\pm$ 2.1	27.9 $\pm$ 0.8	15.7 $\pm$ 1.2	16.3 $\pm$ 3.1	7.7 $\pm$ 2.1	<u>14.0 $\pm$ 1.0</u>	<u>35.7 $\pm$ 1.5</u>	19.4 $\pm$ 8.0
Self-refine	31.0 $\pm$ 3.6	24.3 $\pm$ 0.6	13.3 $\pm$ 3.2	<u>29.7 $\pm$ 2.1</u>	17.6 $\pm$ 1.7	16.0 $\pm$ 3.0	9.0 $\pm$ 2.0	<u>14.0 $\pm$ 1.7</u>	32.0 $\pm$ 2.6	<u>20.8 $\pm$ 8.1</u>
MedPrompt	21.0 $\pm$ 3.5	18.3 $\pm$ 1.5	13.7 $\pm$ 3.2	22.8 $\pm$ 2.1	18.4 $\pm$ 1.3	14.3 $\pm$ 1.2	7.3 $\pm$ 1.5	10.7 $\pm$ 1.5	36.0 $\pm$ 3.0	18.1 $\pm$ 7.9
MedRAG	25.0 $\pm$ 3.5	21.0 $\pm$ 0.9	11.3 $\pm$ 2.3	26.7 $\pm$ 2.5	15.0 $\pm$ 0.6	13.1 $\pm$ 1.6	7.3 $\pm$ 0.6	11.0 $\pm$ 0.0	26.3 $\pm$ 2.3	17.4 $\pm$ 7.0
MultiPersona	30.3 $\pm$ 1.5	21.7 $\pm$ 0.6	16.3 $\pm$ 4.2	26.9 $\pm$ 1.6	16.1 $\pm$ 1.7	15.7 $\pm$ 2.3	6.7 $\pm$ 1.5	11.0 $\pm$ 1.0	35.0 $\pm$ 1.0	20.0 $\pm$ 8.7
MDAgents	24.0 $\pm$ 1.0	21.7 $\pm$ 1.5	17.0 $\pm$ 5.6	20.6 $\pm$ 2.4	14.2 $\pm$ 1.7	12.3 $\pm$ 1.5	5.3 $\pm$ 1.2	7.0 $\pm$ 2.0	32.0 $\pm$ 2.0	17.1 $\pm$ 8.0
AFlow	<u>32.3 $\pm$ 2.1</u>	<u>28.0 $\pm$ 3.0</u>	18.7 $\pm$ 3.5	25.6 $\pm$ 1.6	16.9 $\pm$ 1.2	10.0 $\pm$ 3.0	10.7 $\pm$ 3.5	10.3 $\pm$ 3.5	28.7 $\pm$ 1.5	20.1 $\pm$ 8.3
SPO	17.7 $\pm$ 1.5	20.7 $\pm$ 0.6	25.7 $\pm$ 2.1	22.4 $\pm$ 2.9	22.5 $\pm$ 0.0	15.0 $\pm$ 1.0	11.7 $\pm$ 1.2	13.3 $\pm$ 2.1	32.0 $\pm$ 1.0	20.1 $\pm$ 6.1
MedAgents	25.3 $\pm$ 2.3	22.0 $\pm$ 1.0	12.3 $\pm$ 1.5	24.2 $\pm$ 3.5	18.7 $\pm$ 2.8	12.0 $\pm$ 0.0	7.7 $\pm$ 2.1	10.7 $\pm$ 1.5	31.3 $\pm$ 2.5	18.2 $\pm$ 7.5
MedAgents-X	35.0 $\pm$ 3.6	30.3 $\pm$ 2.1	<u>25.0 $\pm$ 0.0</u>	34.7 $\pm$ 5.0	19.8 $\pm$ 0.6	<u>24.3 $\pm$ 0.6</u>	15.0 $\pm$ 2.6	20.3 $\pm$ 4.0	29.3 $\pm$ 2.1	26.0 $\pm$ 6.5
o3-mini										
Zero-shot	53.0 $\pm$ 2.0	25.3 $\pm$ 1.5	17.3 $\pm$ 1.5	37.0 $\pm$ 1.4	50.9 $\pm$ 1.7	19.3 $\pm$ 1.5	26.3 $\pm$ 1.5	16.3 $\pm$ 1.5	16.3 $\pm$ 1.5	29.1 $\pm$ 13.7
Few-shot	31.7 $\pm$ 0.6	34.3 $\pm$ 2.5	<u>22.0 $\pm$ 3.0</u>	28.8 $\pm$ 1.4	27.3 $\pm$ 0.7	26.3 $\pm$ 1.5	15.7 $\pm$ 1.5	12.7 $\pm$ 2.5	15.7 $\pm$ 4.5	23.8 $\pm$ 7.3
CoT	56.7 $\pm$ 2.1	27.7 $\pm$ 3.2	15.0 $\pm$ 1.7	38.4 $\pm$ 2.8	49.1 $\pm$ 1.3	19.7 $\pm$ 1.5	28.0 $\pm$ 3.6	22.7 $\pm$ 3.2	40.3 $\pm$ 2.1	33.1 $\pm$ 13.2
CoT-SC	55.0 $\pm$ 1.7	26.0 $\pm$ 1.0	14.7 $\pm$ 1.5	37.9 $\pm$ 0.8	50.2 $\pm$ 2.3	18.7 $\pm$ 1.2	28.0 $\pm$ 2.0	22.3 $\pm$ 2.1	38.3 $\pm$ 0.6	32.3 $\pm$ 13.1
Self-refine	52.7 $\pm$ 2.1	27.7 $\pm$ 2.5	17.3 $\pm$ 1.5	36.5 $\pm$ 0.8	50.9 $\pm$ 1.7	22.0 $\pm$ 2.0	<u>32.7 $\pm$ 1.5</u>	22.3 $\pm$ 1.2	31.7 $\pm$ 1.5	32.6 $\pm$ 11.7
MedPrompt	59.7 $\pm$ 1.5	27.3 $\pm$ 1.5	15.3 $\pm$ 1.2	33.8 $\pm$ 2.1	57.3 $\pm$ 4.5	18.3 $\pm$ 1.5	29.7 $\pm$ 0.6	23.3 $\pm$ 3.1	37.3 $\pm$ 0.6	33.6 $\pm$ 14.9
MedRAG	49.7 $\pm$ 4.2	30.3 $\pm$ 2.1	21.3 $\pm$ 2.1	38.4 $\pm$ 4.2	46.9 $\pm$ 1.2	17.3 $\pm$ 3.2	25.7 $\pm$ 1.5	18.3 $\pm$ 1.5	37.7 $\pm$ 2.5	31.7 $\pm$ 11.4
MultiPersona	<u>61.0 $\pm$ 1.0</u>	25.7 $\pm$ 2.1	16.7 $\pm$ 2.9	39.3 $\pm$ 2.9	54.7 $\pm$ 0.7	20.3 $\pm$ 3.1	29.7 $\pm$ 2.9	22.7 $\pm$ 1.5	40.3 $\pm$ 2.1	34.5 $\pm$ 14.6
MDAgents	58.3 $\pm$ 2.1	29.7 $\pm$ 2.1	20.4 $\pm$ 4.6	41.1 $\pm$ 6.3	54.4 $\pm$ 1.5	18.3 $\pm$ 1.5	29.0 $\pm$ 1.0	23.0 $\pm$ 2.6	41.3 $\pm$ 5.0	35.1 $\pm$ 13.7
AFlow	56.3 $\pm$ 1.5	<u>33.0 $\pm$ 2.0</u>	23.3 $\pm$ 1.5	<u>47.9 $\pm$ 1.4</u>	43.9 $\pm$ 1.2	<u>24.7 $\pm$ 1.2</u>	24.0 $\pm$ 2.0	<u>25.3 $\pm$ 1.5</u>	<u>45.3 $\pm$ 1.5</u>	<u>36.0 $\pm$ 11.8</u>
SPO	59.3 $\pm$ 1.5	<u>33.0 $\pm$ 2.0</u>	21.3 $\pm$ 1.5	42.5 $\pm$ 0.7	50.6 $\pm$ 1.2	21.3 $\pm$ 1.5	27.0 $\pm$ 1.0	23.7 $\pm$ 1.5	42.0 $\pm$ 1.0	35.6 $\pm$ 12.9
MedAgents	58.0 $\pm$ 4.0	26.3 $\pm$ 2.5	16.3 $\pm$ 0.6	34.7 $\pm$ 1.6	53.5 $\pm$ 0.7	13.0 $\pm$ 0.0	28.7 $\pm$ 2.5	23.0 $\pm$ 2.6	38.0 $\pm$ 4.0	32.4 $\pm$ 14.6
MedAgents-X	63.0 $\pm$ 1.0	30.0 $\pm$ 2.6	17.3 $\pm$ 2.1	49.3 $\pm$ 2.4	<u>55.4 $\pm$ 1.3</u>	21.7 $\pm$ 2.1	35.0 $\pm$ 2.6	26.3 $\pm$ 1.5	46.3 $\pm$ 2.1	38.3 $\pm$ 15.0

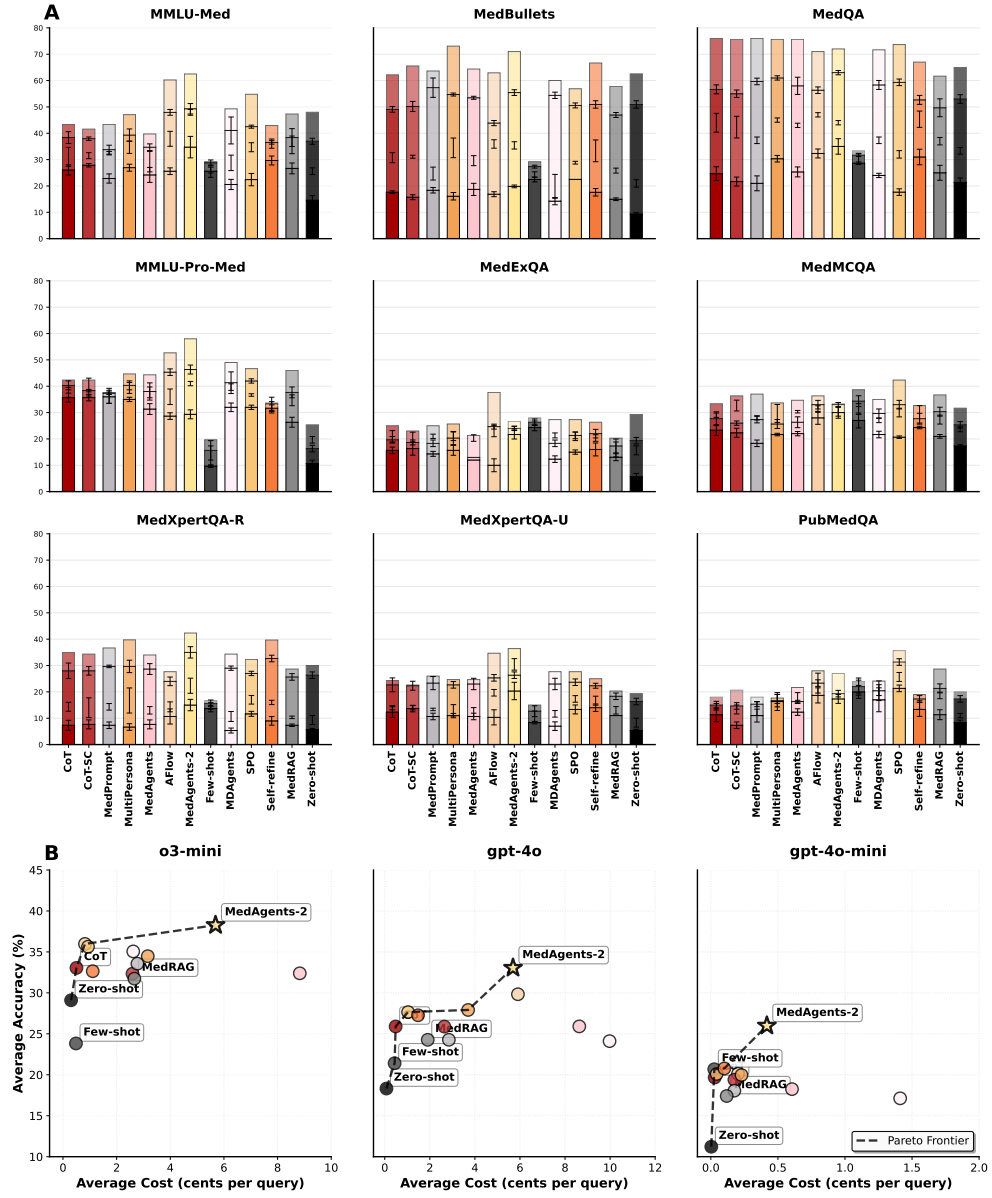


Fig. A1 Performance and efficiency across datasets, models, and methods. a, Accuracy of 13 reasoning approaches on 9 medical QA datasets using GPT-4o, GPT-4o-mini, and o3-mini backbones; bars show mean with variability across runs. b, Accuracy–cost Pareto frontier averaged over datasets; MedAgents-X (star) lies on the frontier, indicating strong cost-effectiveness.

Appendix B The Algorithm

Algorithm 1 outlines the multi-agent evidence-grounded clinical question answering pipeline.

Algorithm 1: Multi-Agent Evidence-Grounded Clinical Question Answering

Input : Clinical case c ; answer candidates $\mathcal{A} = \{a_1, \dots, a_K\}$; static corpora $\mathcal{C} = \{\mathcal{C}_1, \dots, \mathcal{C}_L\}$ with dense retrievers (f_ℓ, g_ℓ) ; web API Web; hyperparameters: retrieval depth k , rerank cutoff m , vote temperature τ , consensus threshold θ , max rounds $R_{\max}$.

Output: Answer $a^* \in \mathcal{A}$; evidence set $\mathcal{E}$; audit trail $\mathcal{T}$.

```

// Phase 1: Difficulty Assessment and Expert Assembly
1  $\gamma \leftarrow \text{Difficulty}(c)$  where  $\gamma \in \{\text{easy}, \text{medium}, \text{hard}\}$ 
2  $(S, \mathbf{w}) \leftarrow \text{Triage}(c, \gamma)$  //  $S = \{s_1, \dots, s_M\}$  experts,  $\mathbf{w} \in \Delta^{M-1}$ 
3 Initialize:  $\mathcal{E} \leftarrow \emptyset, \mathcal{T} \leftarrow \{(\gamma, S, \mathbf{w})\}, r \leftarrow 1$ 

// Phase 2: Iterative Evidence-Based Deliberation
4 while  $r \leq R_{\max}$  do
    // Parallel expert reasoning with evidence retrieval
    for  $i \in \{1, \dots, M\}$  do in parallel
        5  $q_i^{(r)} \leftarrow \text{formulate query from } (c, \mathcal{E}, \mathcal{T})$ 
        6  $\tilde{q}_i^{(r)} \leftarrow \text{Rewrite}(q_i^{(r)}, s_i)$  // domain-aware query expansion
        // Multi-source evidence aggregation
        7  $\mathcal{P}_i^{(r)} \leftarrow \bigcup_{\ell=1}^L \text{Retrieve}(\tilde{q}_i^{(r)}, \mathcal{C}_\ell; k)$  where retrieval uses  $(f_\ell(\tilde{q}_i^{(r)}), g_\ell(d))$ 
        8 if web search enabled then
        9      $\mathcal{P}_i^{(r)} \leftarrow \mathcal{P}_i^{(r)} \cup \text{Retrieve}(\tilde{q}_i^{(r)}, \text{Web}; k)$ 
        10  $\hat{\mathcal{P}}_i^{(r)} \leftarrow \text{Rerank}(q_i^{(r)}, \mathcal{P}_i^{(r)})$  // cross-encoder reranking
        11  $\mathcal{E}_i^{(r)} \leftarrow \text{top-}m(\hat{\mathcal{P}}_i^{(r)})$ 
        12 if review enabled then
        13      $\mathcal{E}_i^{(r)} \leftarrow \text{Review}(\mathcal{E}_i^{(r)}, q_i^{(r)})$  // relevance filtering
        14  $(a_i^{(r)}, \rho_i^{(r)}, \zeta_i^{(r)}) \leftarrow \text{Diagnose}(c, \mathcal{E}_i^{(r)}, s_i)$  //  $a_i^{(r)} \in \mathcal{A}$ : answer,  $\rho_i^{(r)} \in [0, 1]$ : confidence,  $\zeta_i^{(r)}$ : rationale
        // Weighted confidence-aware aggregation
        15 for  $j \in \{1, \dots, K\}$  do
        16      $v_j^{(r)} \leftarrow \sum_{i=1}^M w_i \cdot \rho_i^{(r)} \cdot \mathbb{1}[a_i^{(r)} = a_j]$ 
        17  $p_j^{(r)} \leftarrow \frac{\exp(v_j^{(r)}/\tau)}{\sum_{k=1}^K \exp(v_k^{(r)}/\tau)}$  for  $j \in \{1, \dots, K\}$ 
        18  $\hat{a}^{(r)} \leftarrow \arg \max_j p_j^{(r)}$ 
        // Update global evidence and audit trail
        19  $\mathcal{E} \leftarrow \mathcal{E} \cup \bigcup_{i=1}^M \mathcal{E}_i^{(r)}$ 
        20  $\mathcal{T} \leftarrow \mathcal{T} \cup \{(\{a_i^{(r)}, \rho_i^{(r)}, \zeta_i^{(r)}\}_{i=1}^M, \mathbf{p}^{(r)}, \hat{a}^{(r)})\}$ 
        // Convergence check
        21 if  $\max_j p_j^{(r)} \geq \theta$  or  $r = R_{\max}$  then
        22     break
        // Orchestrator feedback generation
        23  $\mathcal{F}^{(r)} \leftarrow \text{Moderate}(c, \{(a_i^{(r)}, \rho_i^{(r)}, \zeta_i^{(r)})\}_{i=1}^M, \mathbf{p}^{(r)})$ 
        24  $\mathcal{T} \leftarrow \mathcal{T} \cup \{\mathcal{F}^{(r)}\}$ 
        25  $r \leftarrow r + 1$ 
26 return  $(a^* = \hat{a}^{(r)}, \mathcal{E}, \mathcal{T})$ 

```

Appendix C Implementation Details

C.1 Medical Specialties

As is discussed in Section 3, we use a dynamic expert panel to answer medical questions. The expert panel is assembled from a list of medical specialties. The list of medical specialties is shown in Table C2.

Table C2 Full list of medical specialties used to assemble dynamic expert panels in MedAgents-X

Number	Specialist	Number	Specialist
1	Internal Medicine	26	Dermatology
2	Cardiology	27	Pathology
3	Neurology	28	Geriatrics
4	Emergency Medicine	29	Immunology
5	Infectious Disease	30	Rheumatology
6	Pediatrics	31	Urology
7	Endocrinology	32	Obstetrics & Gynecology
8	Pharmacology	33	Surgery
9	Pulmonology	34	Preventive Medicine
10	Gastroenterology	35	Critical Care Medicine
11	Hematology	36	Toxicology
12	Radiology	37	Anesthesiology
13	Nephrology	38	Neurosurgery
14	Psychiatry	39	Family Medicine
15	Oncology	40	Vascular Medicine
16	Genetics	41	Ophthalmology
17	Dermatology	42	Orthopedics
18	Pathology	43	Occupational Medicine
19	Geriatrics	44	Sports Medicine
20	Immunology	45	Public Health
21	Rheumatology	46	Clinical Research
22	Urology	47	Sleep Medicine
23	Obstetrics & Gynecology	48	Allergy & Immunology
24	Surgery	49	Biostatistics
25	Preventive Medicine		

C.2 Prompts

This appendix section documents all prompting templates used in our system. Below we list each prompt for ease of navigation.

Triage Prompts

Difficulty Assessment Prompt

You are a medical education specialist and triage coordinator. Assess the difficulty level of the following medical question based on these criteria:

1. Knowledge depth required (basic facts vs. specialized knowledge)
2. Clinical reasoning complexity (straightforward vs. multi-step reasoning)
3. Ambiguity level (clear vs. nuanced distinctions between options)
4. Specialized terminology or concepts
5. Rarity of the medical condition or treatment discussed

Classify the question as 'easy', 'medium', or 'hard' and provide a brief justification. After determining the difficulty level, you **MUST** transfer to the appropriate triage agent to get the specialty classification and expert profiles. Do not provide the final triage output yourself - you must handoff to the specialized triage agent.

Expert Specialty Triage Prompt

You are a medical question triage expert. Complete the following steps:

1. Read the scenario carefully and analyze the clinical context
2. Classify the question into {n} experts selected from: {medical.specialties}
3. For each selected expert, assign a weight between 0 and 1 (weights must sum to 1)
4. Generate a creative, CV-style expert profile for each specialist including:
 - Name: The name of the expert
 - Past Experience: Detailed summary of clinical or research experience
 - Educational Background: Academic degrees, supervisors, certifications, specialized training
 - Core Specialties: Key areas of expertise with precise descriptions
5. For each expert, provide a unique research focus that distinguishes them from other experts in the same field. This should be a specific area they've published on or have special interest in.

Search Prompts

Medical Evidence Retrieval Prompt

Rewrite the following medical search query to improve document retrieval. Focus particularly on aspects related to {domain} and emphasize terminology specific to this field.

Original query: {query}

Instructions:

1. Expand medical terminology and include synonyms
2. Add relevant anatomical, physiological, or pathological terms
3. Include common abbreviations and full forms
4. Maintain the core meaning while making it more searchable
5. Focus on key medical concepts that would appear in medical literature

For comprehensive analysis, use BOTH search tools when available:

- Use `search_medical_knowledge` first to find established medical knowledge and guidelines
- Use `web_search` to find current research, recent developments, or additional perspectives
- Combining both sources will give you the most complete picture for your analysis

Query Rewrite Prompt

Your previous question '{original_query}' is too similar to questions already asked. Focus on aspects related to {domain}.

Previous queries include: {previous_queries}

Please generate a completely different question that explores a new aspect of the main problem. Ensure your new query addresses a unique angle not covered by these queries.

New query:

Document Review Prompt

Evaluate the relevance of the following document to the given query. Pay special attention to information relevant to {domain}.

Query: {query}

Document: {document}

Please assess:

1. How helpful is this document in answering the query?
2. What specific information from the document is relevant?
3. Rate the relevance from 1-10 (10 being most relevant)
4. Provide a brief justification for your rating

Respond in JSON format with keys: `relevance_score`, `helpfulness`, `relevant_info`, `justification`

Focus on clinical accuracy, evidence quality, and direct applicability to the medical question at hand.

Expert Prompts

Expert Medical Reasoning Prompt

You are {name}, {job_title}.

Your Profile:

- Past Experience: {past_experience}
- Educational Background: {educational_background}
- Core Specialties: {core_specialties}
- Research Focus: {research_focus}

IMPORTANT: You must provide your complete final answer in a single response. Do not ask follow-up questions or request additional information. Use the available tools if needed, then provide your definitive answer.

CONFIDENCE GUIDELINES: Be honest about your confidence level:

- Use 'high' confidence only when you are very certain based on strong evidence or clear clinical guidelines
- Use 'medium' confidence when you have good reasoning but some uncertainty remains
- Use 'low' confidence when the question is outside your expertise, evidence is limited, or multiple answers seem plausible
- It is better to express appropriate uncertainty than to appear overconfident

This is your only opportunity to respond - make it complete and definitive with honest assessment of your confidence.

Expert Final Response Template

Please answer the following medical question using your expertise and the search tools:

Question: {question}

CRITICAL: You must provide your complete final answer in this single interaction. Do not ask for clarification or additional information. Use any available search tools if needed, then immediately provide your comprehensive analysis including:

1. Your thought process and reasoning
2. The most appropriate answer
3. Your honest confidence level in the answer (be realistic about uncertainty)
4. Clear justification for your response

Remember: It is better to express appropriate uncertainty than to appear overconfident. Use 'low' confidence when you have doubts, 'medium' when reasonably sure, and 'high' only when very certain.

This is your only opportunity to respond | make it complete and definitive.

Orchestrator Prompts

Discussion Moderator Prompt

You are an expert medical orchestrator who analyzes responses from medical specialists and provides constructive feedback for multi-round discussions.

Your role is to:

1. Analyze each expert's justification and reasoning
2. Identify strengths and weaknesses in their arguments
3. Provide specific, actionable feedback for improvement
4. Highlight areas of agreement and disagreement
5. Suggest areas for further exploration
6. Evaluate the current decision based on expert responses
7. Decide whether the discussion should continue or if the current consensus is sufficient

When deciding whether to continue:

- Continue if there are significant disagreements that need resolution
- Continue if the confidence in the current decision is low
- Continue if important areas haven't been explored
- Stop if there's strong consensus and high confidence
- Stop if additional rounds are unlikely to improve the decision

Be constructive, specific, and focus on improving the quality of medical reasoning.

Moderator Input Template

Question: {question}

Options:
{formatted_options}

Round {round_index} Expert Responses:
{expert_summaries}

Current Decision Based on Expert Responses:
Final Answer: {final_answer}
Vote Scores: {vote_scores}
Expert Details: {expert_details}

Please analyze these expert responses and the current decision, then provide:

1. A summary of this round
2. Individual feedback for each expert
3. Key insights from the round
4. Areas where experts agree
5. Areas where experts disagree
6. Whether the discussion should continue to another round
7. Your confidence level in the current decision

Focus on the quality of medical reasoning, evidence cited, and whether additional discussion would improve the decision quality.

Appendix D Representative Full Trajectories

We present representative error-category cases in a standardized consultation format. Full prompts, queries, retrieved passages, and citations are captured in the system audit logs.

Reasoning Error.

The panel misinterpreted reliability evidence and chose Schamroth's sign (E) over the ground-truth method (A: inspect nail-bed angle).

Error — Reasoning Error — mmlu-pro / run.0 / o3_mini (idx=6012)

Question: Which is the most reliable way to assess for clubbing?

Options: A: Inspect nail-bed angle from the side; B: Inspect curvature from below; C: Inspect angle from above; D: Nail texture; E: Schamroth's sign; F: Nail width; G: Pain; H: Color; I: Length; J: Fluctuancy.

Ground truth vs. final. Ground truth: A. Final consensus: E.

Triage. Difficulty: easy-medium (physical exam nuance). Meeting: Moderated team discussion. Panel size: 3. Rounds: 1.

Panel. Dr. Sara Kim (Pulmonology, $w=0.4$); Dr. James O'Neill (Cardiology, $w=0.3$); Dr. Mia Rossi (Internal Medicine, $w=0.3$).

Moderator agenda. (1) Identify most reliable bedside test; (2) Contrast Schamroth's sign vs nail-bed angle reliability; (3) Cite an authoritative exam source.

Round 1 agenda. Compare reliability statements across exam methods and select a single best answer.

- Dr. Sara Kim (Pulmonology) — vote: E (high). Query: "clubbing assessment reliability schamroth vs lovibond nail bed angle most reliable"; Retrieved (highlights): physical diagnosis texts calling Schamroth a classic test; summaries equating Schamroth with confirmation of clubbing. Rationale: Schamroth is the recognized clinical sign.
- Dr. James O'Neill (Cardiology) — vote: E (high). Query: "digital clubbing physical exam reliability schamroth sign vs nail bed angle"; Retrieved (highlights): reviews emphasizing Schamroth's window loss; mention of increased Lovibond angle. Rationale: Schamroth most reliable.
- Dr. Mia Rossi (Internal Medicine) — vote: E (high). Query: "most reliable method assess clubbing bedside schamroth sign"; Retrieved (highlights): teaching resources labeling Schamroth as most reliable; limited discussion of angle inspection reliability. Rationale: Select E.

Moderator action: Stop. Endorses unanimous E; does not challenge the distinction between "classic sign" and "most reliable method."

Consensus and outcome. Final: E (incorrect). *Notes.* Team overweighted "classic sign" language; benchmark specifies direct reliability of the side-view nail-bed angle (A).

Audit trail elements. Full queries, retrieved passages, and citations are logged; this excerpt shows representative retrieval summaries only.

Evidence Error.

Evidence cited supported topical estrogen, but the benchmark keyed on nonhormonal first-line (vaginal moisturizers), leading to a mismatch.

Error — Evidence Error — medbullets / run_2 / o3_mini (idx=116)

Question: 59-year-old postmenopausal woman with urinary urgency, dyspareunia, and atrophic exam findings; most appropriate next step?

Options: A: Metronidazole; B: Nitrofurantoin; C: Topical clobetasol; D: Topical estrogen; E: Vaginal moisturizers.

Ground truth vs. final. Ground truth: E. Final consensus: D.

Triage. Difficulty: medium (GSM management nuances). Meeting: Moderated team discussion. Panel size: 3. Rounds: 1.

Panel. Dr. Priya Shah (Ob/Gyn, $w=0.4$); Dr. Elena Morales (Urogynecology, $w=0.3$); Dr. Daniel Wu (Internal Medicine, $w=0.3$).

Moderator agenda. (1) Identify guideline-backed first step for GSM without vasomotor symptoms; (2) Contrast hormonal vs nonhormonal options under benchmark conventions; (3) Provide citations.

Round 1 agenda. Map exam to GSM and select initial therapy.

- Dr. Priya Shah (Ob/Gyn) — vote: D (high). Query: “genitourinary syndrome of menopause first line therapy topical estrogen moisturizers guidelines”; Retrieved (highlights): texts emphasizing local estrogen efficacy; limited mention of moisturizers as trial option. Rationale: Start topical estrogen.
- Dr. Elena Morales (Urogynecology) — vote: D (high). Query: “atrophic vaginitis postmenopausal urinary urgency local estrogen first line”; Retrieved (highlights): summaries supporting vaginal estrogen for epithelial restoration. Rationale: Choose estrogen.
- Dr. Daniel Wu (Internal Medicine) — vote: D (high). Query: “GSM management algorithm moisturizers vs estrogen first line”; Retrieved (highlights): sources prioritizing estrogen for symptomatic GSM. Rationale: Estrogen best addresses pathophysiology.

Moderator action: Stop. Accepts D; does not reconcile benchmark’s nonhormonal-first convention (E) for this item.

Consensus and outcome. Final: D (incorrect). *Notes.* Evidence selection favored hormonal efficacy while the dataset keyed on nonhormonal moisturizers as initial step, creating an evidence–label mismatch.

Audit trail elements. Full queries, retrieved passages, and citations are logged; this excerpt shows representative retrieval summaries only.

Misled Discussion.

Experts asserted superiority of AFC with high confidence despite the study’s framing calling for a nuanced “maybe” (GT=C).

Question: Does baseline antral follicle assessment better predict IVF outcome than age and FSH?

Options: A: yes; B: no; C: maybe.

Ground truth vs. final. Ground truth: C. Final consensus: A.

Triage. Difficulty: medium (single-study inference). Meeting: Moderated team discussion. Panel size: 3. Rounds: 1.

Panel. Dr. Noor Al-Mansouri (Reproductive Endocrinology, $w=0.4$); Dr. Kevin Liu (Biostatistics, $w=0.3$); Dr. Anna Petrov (Ob/Gyn, $w=0.3$).

Moderator agenda. (1) Summarize study findings; (2) Evaluate generalizability and uncertainty; (3) Choose between categorical vs qualified answer.

Round 1 agenda. Interpret study outcomes controlling for age/FSH and determine answer framing.

- Dr. Noor Al-Mansouri (Repro Endo) — vote: A (high). Query: “antral follicle count predictive IVF outcomes vs age FSH study pregnancy rates cancellation”; Retrieved (highlights): AFC>6 shows higher pregnancy rate (51% vs 19%) and lower cancellations. Rationale: AFC better predictor.
- Dr. Kevin Liu (Biostatistics) — vote: A (high). Query: “AFC independent predictor IVF controlling for age FSH significance”; Retrieved (highlights): significance maintained after adjustment; limited external validity discussion. Rationale: Conclude superiority.
- Dr. Anna Petrov (Ob/Gyn) — vote: A (high). Query: “baseline antral follicle assessment better than age FSH IVF outcome”; Retrieved (highlights): narrative emphasizing AFC utility; no explicit caveats. Rationale: Support yes.

Moderator action: Stop. Accepts unanimous A; uncertainty framing (*maybe*) not considered despite preliminary study design.

Consensus and outcome. Final: A (incorrect). Notes. Overconfident interpretation of a preliminary single-center study; benchmark expects qualified *maybe* (C).

Audit trail elements. Full queries, retrieved passages, and citations are logged; this excerpt shows representative retrieval summaries only.

Insufficient Discussion.

Discussion stopped with a three-way tie (A/C/E) and unresolved contrasts among lab sets; consensus not reached.

Question: 69-year-old with dyspnea, S3, crackles, edema; which lab set (A–E) best matches CHF?

Ground truth vs. final. Ground truth: C. Final selection at stop: A (with unresolved tie across rounds).

Triage. Difficulty: medium (pattern matching vs numeric nuance). Meeting: Moderated team discussion. Panel size: 3. Rounds: 3.

Panel. Dr. Robert Chen (Cardiology, $w=0.4$); Dr. Lila Desai (Nephrology, $w=0.3$); Dr.

Marco Alvarez (Internal Medicine, $w=0.3$).

Moderator agenda. (1) Confirm CHF profile: elevated BNP, dilutional hyponatremia, possible renal indices; (2) Perform side-by-side comparison of A/C/E; (3) Resolve to a single choice.

Round 1 agenda. Map clinical picture to typical lab abnormalities and pick a candidate set.

- Dr. Robert Chen (Cardiology) — vote: A (high). Query: “CHF labs typical BNP sodium bun creatinine pattern option sets”; Retrieved (highlights): BNP elevation; hyponatremia common; variable renal indices. Rationale: A fits.
- Dr. Lila Desai (Nephrology) — vote: C (high). Query: “decompensated heart failure lab abnormalities hyponatremia bun creatinine ratio”; Retrieved (highlights): dilutional hyponatremia and prerenal pattern; prefers C. Rationale: Choose C.
- Dr. Marco Alvarez (Internal Med) — vote: E (high). Query: “CHF decompensated expected labs which set best matches”; Retrieved (highlights): general CHF profile; argues E aligns. Rationale: Select E.

Moderator action: Continue. Requests explicit numeric comparison among A/C/E.

Round 2 agenda. Contrast sodium levels, BUN/Cr ratios, and BNP magnitude across A/C/E.

- Dr. Robert Chen — vote: A (high). Retrieved: examples where BNP notably high and sodium mildly low; A emphasized. Rationale unchanged.
- Dr. Lila Desai — vote: C (high). Retrieved: prerenal indices with hyponatremia; maintains C. Rationale unchanged.
- Dr. Marco Alvarez — vote: E (high). Retrieved: sources equating E with typical CHF; no numeric table provided. Rationale unchanged.

Moderator action: Continue. Asks for a side-by-side table; none produced.

Round 3 agenda. Attempt final reconciliation and rule-out of non-matching sets.

- Dr. Robert Chen — vote: A (high). Rationale unchanged.
- Dr. Lila Desai — vote: C (high). Rationale unchanged.
- Dr. Marco Alvarez — vote: E (high). Rationale unchanged.

Moderator action: Stop (premature). Declares A based on perceived decisiveness despite three-way split; unresolved contrasts remain.

Consensus and outcome. Final: A (incorrect). *Notes.* Failure to execute a required side-by-side numeric comparison resulted in premature termination without convergence; benchmark expects C.

Audit trail elements. Full queries, retrieved passages, and citations are logged; this excerpt shows representative retrieval summaries only.

Misled Discussion.

A benchmark-aligned alternative was not adopted: although the panel argued that FINDER excludes thumbs, the ground truth keyed on little finger (C), and that view was not incorporated into the final choice.

Question: Fingerprinting (FINDER) records 8 fingers. Which pair is excluded?

Options: A: Ring; B: Thumb; C: Little; D: Middle.

Ground truth vs. final. Ground truth: C. Final consensus: B.

Triage. Difficulty: easy (protocol fact). Meeting: Moderated team discussion. Panel size: 3. Rounds: 1.

Panel. Dr. Hannah Brooks (Forensic Medicine, $w=0.4$); Dr. Omar Haddad (Pathology, $w=0.3$); Dr. Linh Tran (Emergency Medicine, $w=0.3$).

Moderator agenda. (1) Identify standard 8-finger protocol; (2) cite a forensics reference; (3) resolve excluded pair.

Round 1 agenda. State which fingers are recorded in an 8-finger capture and select the excluded pair.

- Dr. Hannah Brooks (Forensics) — vote: B (high). Query: “FINDER fingerprinting records 8 fingers which pair excluded thumbs”; Retrieved (highlights): practice summaries listing index/middle/ring/little from both hands. Rationale: Exclude thumbs.
- Dr. Omar Haddad (Pathology) — vote: B (high). Query: “8 finger fingerprint protocol exclude thumbs standard”; Retrieved (highlights): general protocols excluding thumbs. Rationale: Choose thumbs.
- Dr. Linh Tran (Emergency Med) — vote: C (high). Query: “FINDER 8 finger which excluded thumbs vs little”; Retrieved (highlights): examples excluding thumbs; no mention of little finger exclusion. Rationale: Select C.

Moderator action: Stop. Accepts B; does not surface benchmark-aligned alternative (C) despite unanimity being based on secondary sources.

Consensus and outcome. Final: B (incorrect). *Notes.* A valid alternative answer (C) aligned with the benchmark was ignored; unanimity masked a systematic assumption.

Audit trail elements. Full queries, retrieved passages, and citations are logged; this excerpt shows representative retrieval summaries only.